
Publication List

493 publications · Chunhua Shen

- 493. Z. Luo, Z. Du, Z. Zhu, H. Zhong, M. Zhu, W. Wang, Y. Xi, C. Jing, H. Chen, [C. Shen](#) (2026), “[Preserving Source Video Realism: High-Fidelity Face Swapping for Cinematic Quality](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 492. M. Zhu, S. Jiang, H. Zheng, Z. Luo, H. Zhong, A. Li, K. Wang, J. Rong, Y. Liu, H. Chen, T. Lin, [C. Shen](#) (2026), “[Exploring Spatial Intelligence From a Generative Perspective](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 491. G. Xu, H. Geng, H. Zheng, S. Yin, Y. Sun, H. Chen, [C. Shen](#) (2026), “[Unlocking the Power of Critical Factors for 3D Visual Geometry Estimation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 490. H. Zhong, M. Zhu, S. Zeng, A. Li, C. Chen, H. Geng, D. Shi, W. Ye, T. Lin, H. Chen, [C. Shen](#) (2026), “[Eliciting Complex Spatial Reasoning in MLLMs Through Wide-Baseline Matching](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 489. S. Shen, K. Liu, J. Xie, S. Gao, [C. Shen](#), G. Liu, M. Crispin-Ortuzar, S. Gao (2026), “[R²-Seg: Training-Free OOD Medical Tumor Segmentation Via Anatomical Reasoning and Statistical Rejection](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 488. M. Liu, J. Shu, H. Chen, Z. Li, C. Zhao, J. Yang, S. Gao, H. Chen, [C. Shen](#) (2026), “[StaMo: Unsupervised Learning of Generalizable Robot Motion From Compact State Representation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 487. X. He, P. Lin, Y. Cui, D. Guo, [C. Shen](#), X. Zhang (2026), “[GeoMotion: Rethinking Motion Segmentation Via Latent 4D Geometry](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'26)*.
- 486. M. Zhu, H. Zhong, C. Zhao, Z. Du, Z. Huang, M. Liu, H. Chen, C. Zou, J. Chen, M. Yang, [C. Shen](#) (2026), “[ACTIVE-O3: Empowering Multimodal Large Language Models With Active Perception Via GRPO](#)”, *Int. Conf. Machine Learning (ICML'26)*.
- 485. M. Liu, Z. Huang, X. Lin, M. Zhu, C. Zhao, Y. Wang, H. Zhu, H. Chen, [C. Shen](#) (2026), “[GAE: Unleashing Physical Potential of VLM With Generalizable Action Expert](#)”, *Int. Conf. Machine Learning (ICML'26)*.
- 484. G. Ding, X. Xu, H. Chen, [C. Shen](#) (2026), “[SpeedVFI: One-Step Diffusion for Efficient Video Frame Interpolation](#)”, *Int. Conf. Machine Learning (ICML'26)*.
- 483. H. Deng, T. Pan, F. Zhang, Y. Liu, Z. Luo, Y. Cui, W. Wang, [C. Shen](#), S. Shan, Z. Zhang, X. Wang (2026), “[Uniform Discrete Diffusion With Metric Path for Video Generation](#)”, *Int. Conf. Learning Representations (ICLR'26)*.
- 482. Y. Wang, J. Zhou, H. Zhu, W. Chang, Y. Zhou, Z. Li, J. Chen, J. Pang, [C. Shen](#), T. He (2026), “ [\$\pi^3\$: Permutation-Equivariant Visual Geometry Learning](#)”, *Int. Conf. Learning Representations (ICLR'26)*.
- 481. Y. Zhou, Y. Wang, J. Zhou, W. Chang, H. Guo, Z. Li, K. Ma, X. Li, Y. Wang, H. Zhu, M. Liu, D. Liu, J. Yang, Z. Fu, J. Chen, [C. Shen](#), J. Pang, K. Zhang, T. He (2026), “[OmniWorld: A Multi-Domain and Multi-Modal Dataset for 4D World Modeling](#)”, *Int. Conf. Learning Representations (ICLR'26)*.
- 480. Z. Li, J. Zhou, Y. Wang, H. Guo, W. Chang, Y. Zhou, H. Zhu, J. Chen, [C. Shen](#), T. He (2026), “[WinT3R: Window-Based Streaming Reconstruction With Camera Token Pool](#)”, *Int. Conf. Learning Representations (ICLR'26)*.
- 479. W. Wang, B. Fang, C. Jing, Y. Shen, Y. Shen, Q. Wang, H. Ouyang, H. Chen, [C. Shen](#) (2026), “[Time Is a Feature: Exploiting Temporal Dynamics in Diffusion Language Models](#)”, *Int. Conf. Learning Representations (ICLR'26)*.

478. C. Zhao, X. Li, T. Feng, Z. Zhao, H. Chen, [C. Shen](#) (2026), “[TINKER: Diffusion'S Gift to 3D–Multi-View Consistent Editing From Sparse Inputs Without per-Scene Optimization](#)”, *Int. Conf. Learning Representations (ICLR'26)*.
477. K. Wang, L. Lu, M. Liu, J. Jiang, Z. Li, B. Zhang, W. Zheng, X. Yu, H. Chen, [C. Shen](#) (2026), “[ODYSSEY: Open-World Quadrupeds Exploration and Manipulation for Long-Horizon Tasks](#)” [Oral presentation], *AAAI Conf. Artificial Intelligence (AAAI'26)*.
476. C. Jing, H. Zhang, J. Lu, Y. Liu, H. Chen, X. Zhang, [C. Shen](#) (2026), “[Multi-Modal Primitive Retrieval for Compositional Zero-Shot Learning](#)”, *Int. J. Computer Vision*.
475. D. Ma, X. Qi, C. Ye, J. Liang, S. Wen, Y. Li, K. Ding, Y. Hao, J. Fei, W. Mao, L. Li, Z. Lin, Y. Shen, H. Zhu, Y. Hu, R. Zhang, P. Ji, Y. Lu, B. Liu, H. Wang, Y. Chen, Z. Ma, P. Yang, X. Xu, J. Wu, Y. Zhu, Q. Zou, W. Zhu, K. Yao, S. Li, H. Xin, D. Ergu, J. Zeng, Z. Xiao, [C. Shen](#) (2026), “[Atlas of Predicted Protein Complex Structures Across Kingdoms](#)”, *Nature Communications*.
474. W. Wu, M. Liu, Z. Zhu, F. Haoen, X. Xia, W. Wang, K.Q. Lin, [C. Shen](#), M.Z. Shou (2025), “[MovieBench: A Hierarchical Movie Level Dataset for Long Video Generation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'25)*.
473. M. Zhu, Y. Tian, H. Chen, C. Zhou, Q. Guo, Y. Liu, M. Yang, [C. Shen](#) (2025), “[SegAgent: Exploring Pixel Understanding Capabilities in MLLMs by Imitating Human Annotator Trajectories](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'25)*.
472. Y. Liu, Y. Yin, C. Jin, M. Zhu, H. Chen, Y. Xi, B. Feng, H. Wang, S. Li, [C. Shen](#) (2025), “[Unified Open-World Segmentation With Multi-Modal Prompts](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
471. Z. Gao, J. Bian, G. Lin, H. Chen, [C. Shen](#) (2025), “[SurfaceSplat: Connecting Surface Reconstruction and Gaussian Splatting](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
470. H. Zhu, Y. Wang, J. Zhou, W. Chang, Y. Zhou, Z. Li, J. Chen, [C. Shen](#), J. Pang, T. He (2025), “[Aether: Geometric-Aware Unified World Modeling](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
469. S. Zhang, Y. Ge, J. Tian, G. Xu, H. Chen, C. Lv, [C. Shen](#) (2025), “[POMATO: Marrying Pointmap Matching With Temporal Motions for Dynamic 3D Reconstruction](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
468. S. Chan, Z. Li, X. Zhang, W. Li, S. Lu, [C. Shen](#) (2025), “[SMSTracker: Tri-Path Score Mask Sigma Fusion for Multi-Modal Tracking](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
467. J. Zhu, Y. Ong, [C. Shen](#), G. Pang (2025), “[Fine-Grained Abnormality Prompt Learning for Zero-Shot Anomaly Detection](#)”, *IEEE Int. Conf. Computer Vision (ICCV'25)*.
466. C. Zhao, Y. Sun, M. Liu, H. Zheng, M. Zhu, Z. Zhao, H. Chen, T. He, [C. Shen](#) (2025), “[DICEPTION: A Generalist Diffusion Model for Visual Perceptual Tasks](#)” [Spotlight presentation], *Advances in Neural Information Processing Systems (NeurIPS'25)*.
465. H. Zhong, M. Zhu, Z. Du, Z. Huang, C. Zhao, M. Liu, W. Wang, H. Chen, [C. Shen](#) (2025), “[Omni-R1: Reinforcement Learning for Omnimodal Reasoning Via Two-System Collaboration](#)”, *Advances in Neural Information Processing Systems (NeurIPS'25)*.
464. K. Liu, H. Chen, [C. Shen](#) (2025), “[Physics Aware Neural Networks for Unsupervised Binding Energy Prediction](#)”, *Int. Conf. Machine Learning (ICML'25)*.
463. H. Yang, D. Huang, W. Yin, [C. Shen](#), H. Liu, X. He, B. Lin, W. Ouyang, T. Hi (2025), “[Depth Any Video With Scalable Synthetic Data](#)”, *Int. Conf. Learning Representations (ICLR'25)*.
462. X. Jiao, W. Mao, W. Jin, P. Yang, H. Chen, [C. Shen](#) (2025), “[Boltzmann-Aligned Inverse Folding Model as a Predictor of Mutational Effects on Protein-Protein Interactions](#)” [Spotlight presentation], *Int. Conf. Learning Representations (ICLR'25)*.
461. Y. Bo, W. Mao, Y. Shao, W. Bai, P. Ye, X. Ma, J. Zhao, H. Chen, [C. Shen](#) (2025), “[Revisiting Convolution Architecture in the Realm of DNA Foundation Models](#)”, *Int. Conf. Learning Representations (ICLR'25)*.

460. C. Chen, M. Liu, C. Jing, Y. Zhou, F. Rao, H. Chen, B. Zhang, C. Shen (2025), “[PerturboLLaVA: Reducing Multimodal Hallucinations With Perturbative Visual Training](#)”, *Int. Conf. Learning Representations (ICLR'25)*.
459. W. Wang, Q. Wang, K. Zheng, H. Ouyang, Z. Chen, B. Gong, H. Chen, Y. Shen, C. Shen (2025), “[Framer: Interactive Frame Interpolation](#)”, *Int. Conf. Learning Representations (ICLR'25)*.
458. C. Zhao, M. Liu, W. Wang, W. Chen, F. Wang, H. Chen, B. Zhang, C. Shen (2025), “[MovieDreamer: Hierarchical Generation for Coherent Long Visual Sequences](#)”, *Int. Conf. Learning Representations (ICLR'25)*.
457. G. Xu, Y. Ge, M. Liu, C. Fan, K. Xie, Z. Zhao, H. Chen, C. Shen (2025), “[What Matters When Repurposing Diffusion Models for General Dense Perception Tasks?](#)”, *Int. Conf. Learning Representations (ICLR'25)*.
456. Y. Ge, K. Xie, G. Xu, M. Liu, L. Ke, L. Huang, H. Xue, H. Chen, C. Shen (2025), “[Generative Video Matting](#)”, *ACM Special Interest Group on Computer Graphics & Interactive Techniques Conference (SIGGRAPH'25)*.
455. D. Yan, P. Li, Y. Li, H. Chen, Q. Chen, W. Luo, W. Dong, Q. Yan, H. Zhang, C. Shen (2025), “[TG-LLaVA: Text Guided LLaVA Via Learnable Latent Embeddings](#)”, *AAAI Conf. Artificial Intelligence (AAAI'25)*.
454. Q. Wang, L. Liu, C. Jing, P. Wang, Y. Zhang, C. Shen (2025), “[Leaning Dual-Stream Conditional Concepts in Compositional Zero-Shot Learning](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
453. Y. Ge, W. Wang, Y. Chen, F. Wang, L. Yang, H. Chen, C. Shen (2025), “[Diffusion Models Are Efficient Data Generators for Human Mesh Recovery](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
452. H. Zhu, H. Yang, X. Wu, D. Huang, S. Zhang, X. He, H. Zhao, C. Shen, Y. Qiao, T. He, W. Ouyang (2025), “[PonderV2: Pave the Way for 3D Foundation Model With a Universal Pre-Training Paradigm](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
451. Y. Liu, M. Huang, H. Yan, L. Deng, W. Wu, H. Lu, C. Shen, L. Jin, X. Bai (2025), “[VimTS: A Unified Video and Image Text Spotter for Enhancing the Cross-Domain Generalization](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
450. C. Zhao, G. Ding, W. Wang, Z. Yang, Z. Liu, H. Chen, C. Shen (2025), “[FreeCustom: Training-Free Multi-Concept Customization for Image and Video Generation](#)”, *Int. J. Computer Vision*.
449. Y. Liu, M. Zhu, H. Chen, X. Wang, B. Feng, H. Wang, S. Li, R. Vemulapalli, C. Shen (2025), “[Segment Anything in Context With Vision Foundation Models](#)”, *Int. J. Computer Vision*.
448. W. Wu, Z. Li, Y. He, M.Z. Shou, C. Shen, L. Cheng, Y. Li, T. Gao, Z. Di (2025), “[Paragraph-to-Image Generation With Information-Enriched Diffusion Model](#)”, *Int. J. Computer Vision*.
447. P. Jiang, Y. Yang, Y. Cao, Q. Hou, M. Cheng, C. Shen (2024), “[Traffic Scene Parsing Through the TSP6K Dataset](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'24)*.
446. C. Fan, M. Zhu, H. Chen, Y. Liu, W. Wu, H. Zhang, C. Shen (2024), “[DiverGen: Improving Instance Segmentation by Learning Wider Data Distribution With More Diverse Generative Data](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'24)*.
445. G. Ding, C. Zhao, W. Wang, Z. Yang, Z. Liu, H. Chen, C. Shen (2024), “[FreeCustom: Tuning-Free Customized Image Generation for Multi-Concept Composition](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'24)*.
444. Z. Chen, W. Wang, Z. Yang, Z. Yuan, H. Chen, C. Shen (2024), “[FreeCompose: Generic Zero-Shot Image Composition With Diffusion Prior](#)”, *European Conf. Computer Vision (ECCV'24)*.
443. X. Chu, J. Su, B. Zhang, C. Shen (2024), “[VisionLLaMA: A Unified LLaMA Backbone for Vision Tasks](#)”, *European Conf. Computer Vision (ECCV'24)*.
442. Y. Liu, C. Jing, H. Li, M. Zhu, H. Chen, X. Wang, C. Shen (2024), “[A Simple Image Segmentation Framework Via in-Context Examples](#)”, *Advances in Neural Information Processing Systems (NeurIPS'24)*.

441. M. Zhu, Y. Liu, Z. Luo, C. Jing, H. Chen, G. Xu, X. Wang, [C. Shen](#) (2024), “[Unleashing the Potential of the Diffusion Model in Few-Shot Semantic Segmentation](#)”, *Advances in Neural Information Processing Systems (NeurIPS'24)*.
440. D. Chen, Z. Zhou, C. Wang, [C. Shen](#), S. Lyu (2024), “[On the Trajectory Regularity of ODE-Based Diffusion Sampling](#)”, *Int. Conf. Machine Learning (ICML'24)*.
439. K. Liu, S. Shen, W. Mao, X. Jiao, Z. Sun, H. Chen, [C. Shen](#) (2024), “[Floating Anchor Diffusion Model for Multi-Motif Scaffolding](#)”, *Int. Conf. Machine Learning (ICML'24)*.
438. M. Zhu, C. Fan, H. Chen, Y. Liu, W. Mao, X. Xu, [C. Shen](#) (2024), “[Generative Active Learning for Long-Tailed Instance Segmentation](#)”, *Int. Conf. Machine Learning (ICML'24)*.
437. W. Mao, M. Zhu, Z. Sun, S. Shen, L.Y. Wu, H. Chen, [C. Shen](#) (2024), “[De Novo Protein Design Using Geometric Vector Field Networks](#)” [[Spotlight presentation](#)], *Int. Conf. Learning Representations (ICLR'24)*.
436. Y. Liu, M. Zhu, H. Li, H. Chen, X. Wang, [C. Shen](#) (2024), “[Matcher: Segment Anything With One Shot Using All-Purpose Feature Matching](#)”, *Int. Conf. Learning Representations (ICLR'24)*.
435. Z. Yang, D. Gui, W. Wang, H. Chen, B. Zhuang, [C. Shen](#) (2024), “[Object-Aware Inversion and Reassembly for Image Editing](#)”, *Int. Conf. Learning Representations (ICLR'24)*.
434. J. Wang, Y. Cui, D. Guo, J. Li, Q. Liu, [C. Shen](#) (2024), “[PointAttN: You Only Need Attention for Point Cloud Completion](#)”, *AAAI Conf. Artificial Intelligence (AAAI'24)*.
433. C. Jing, Y. Li, H. Chen, [C. Shen](#) (2024), “[Retrieval-Augmented Primitive Representations for Compositional Zero-Shot Learning](#)”, *AAAI Conf. Artificial Intelligence (AAAI'24)*.
432. Y. Li, C. Zhang, G. Yu, Z. Wang, B. Fu, G. Lin, [C. Shen](#), L. Chen, Y. Wei (2024), “[StableLLaVA: Enhanced Visual Instruction Tuning With Synthesized Image-Dialogue Data](#)”, *Findings of the Association for Computational Linguistics (EACL'24)*.
431. M. Zhang, H. Chen, [C. Shen](#), Z. Yang, L. Ou, X. Yu, B. Zhuang (2024), “[LoRAPrune: Structured Pruning Meets Low-Rank Parameter-Efficient Fine-Tuning](#)”, *Findings of the Association for Computational Linguistics (EACL'24)*.
430. M. Hu, W. Yin, C. Zhang, Z. Cai, X. Long, H. Chen, K. Wang, G. Yu, [C. Shen](#), S. Shen (2024), “[Metric3D V2: A Versatile Monocular Geometric Foundation Model for Zero-Shot Metric Depth and Surface Normal Estimation](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
429. R. Li, C. Zhang, Z. Wang, [C. Shen](#), G. Lin (2024), “[Self-Supervised 3D Scene Flow Estimation and Motion Prediction Using Local Rigidity Prior](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
428. W. Wang, C. Zhao, H. Chen, Z. Chen, K. Zheng, [C. Shen](#) (2024), “[AutoStory: Generating Diverse Storytelling Images With Minimal Human Effort](#)”, *Int. J. Computer Vision*.
427. W. Wu, [C. Shen](#), Y. Cai, D. Zhang, Y. Fu, P. Luo, H. Zhou (2024), “[End-to-End Video Text Spotting With Transformer](#)”, *Int. J. Computer Vision*.
426. W. Yin, Y. Liu, [C. Shen](#), B. Sun, A. van den Hengel (2024), “[Scaling up Multi-Domain Semantic Segmentation With Sentence Embeddings](#)”, *Int. J. Computer Vision*.
425. Y. Liu, X. Wang, M. Zhu, Y. Cao, T. Huang, [C. Shen](#) (2024), “[Masked Channel Modeling for Bootstrapping Visual Pre-Training](#)”, *Int. J. Computer Vision*.
424. K. Xian, Z. Cao, [C. Shen](#), G. Lin (2024), “[Towards Robust Monocular Depth Estimation: A New Baseline and Benchmark](#)”, *Int. J. Computer Vision*.
423. H. Li, J. Hu, B. Li, H. Chen, Y. Zheng, [C. Shen](#) (2024), “[Target Before Shooting: Accurate Anomaly Detection and Localization Under One Millisecond Via Cascade Patch Retrieval](#)”, *IEEE Trans. Image Processing*.
422. Q. Wang, L. Liu, C. Jing, H. Chen, G. Liang, P. Wang, [C. Shen](#) (2023), “[Learning Conditional Attributes for Compositional Zero-Shot Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'23)*.

421. X. Wang, W. Wang, Y. Cao, C. Shen, T. Huang (2023), “[Images Speak in Images: A Generalist Painter for in-Context Visual Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'23)*.
420. G. Xu, W. Yin, H. Chen, C. Shen, K. Cheng, F. Zhao (2023), “[Pose-Free 3D Scene Reconstruction With Frozen Depth Models](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
419. Y. Zhao, Q. Ye, W. Wu, C. Shen, F. Wan (2023), “[Generative Prompt Model for Weakly Supervised Object Localization](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
418. W. Yin, C. Zhang, H. Chen, Z. Cai, G. Yu, K. Wang, X. Chen, C. Shen (2023), “[Metric3D: Towards Zero-Shot Metric 3D Prediction From a Single Image](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
417. C. Zhang, W. Yin, G. Yu, Z. Wang, T. Chen, B. Fu, J.T. Zhou, C. Shen (2023), “[Robust Geometry-Preserving Depth Estimation Using Differentiable Rendering](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
416. M. Zhu, H. Li, H. Chen, C. Fan, W. Mao, C. Jing, Y. Liu, C. Shen (2023), “[SegPrompt: Boosting Open-World Segmentation Via Category-Level Prompt Learning](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
415. K. Ying, Q. Zhong, W. Mao, Z. Wang, H. Chen, L.Y. Wu, Y. Liu, C. Fan, Y. Zhuge, C. Shen (2023), “[CTVIS: Consistent Training for Online Video Instance Segmentation](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
414. X. Wang, X. Zhang, Y. Cao, W. Wang, C. Shen, T. Huang (2023), “[SegGPT: Towards Segmenting Everything in Context](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
413. W. Wu, Y. Zhao, M.Z. Shou, H. Zhou, C. Shen (2023), “[Diffumask: Synthesizing Images With Pixel-Level Annotations for Semantic Segmentation Using Diffusion Models](#)”, *IEEE Int. Conf. Computer Vision (ICCV'23)*.
412. W. Wang, Y. Ge, H. Mei, Z. Cai, Q. Sun, C. Shen, Y. Wang, L. Yang, T. Komura (2023), “[Zolly: Zoom Focal Length Correctly for Perspective-Distorted Human Mesh Reconstruction](#)” [Oral presentation], *IEEE Int. Conf. Computer Vision (ICCV'23)*.
411. W. Wu, Y. Zhao, H. Chen, Y. Gu, R. Zhao, Y. He, H. Zhou, M.Z. Shou, C. Shen (2023), “[DatasetDM: Synthesizing Data With Perception Annotations Using Diffusion Models](#)”, *Advances in Neural Information Processing Systems (NeurIPS'23)*.
410. X. Chu, Z. Tian, B. Zhang, X. Wang, X. Wei, H. Xia, C. Shen (2023), “[Conditional Positional Encodings for Vision Transformers](#)”, *Int. Conf. Learning Representations (ICLR'23)*.
409. G. Pang, C. Shen, H. Jin, A. van den Hengel (2023), “[Deep Weakly-Supervised Anomaly Detection](#)”, *ACM SIGKDD Conf. Knowledge Discovery & Data Mining (KDD'23)*.
408. Y. Qin, X. Chen, C. Chen, Y. Shen, B. Ren, Y. Gu, J. Yang, C. Shen (2023), “[FoPro: Few-Shot Guided Robust Webly-Supervised Prototypical Learning](#)”, *AAAI Conf. Artificial Intelligence (AAAI'23)*.
407. Y. Ge, Q. Zhou, X. Wang, Z. Wang, H. Li, C. Shen (2023), “[Point-Teaching: Weakly Semi-Supervised Object Detection With Point Annotations](#)”, *AAAI Conf. Artificial Intelligence (AAAI'23)*.
406. L. Sun, J. Bian, H. Zhan, W. Yin, I. Reid, C. Shen (2023), “[SC-DepthV3: Robust Self-Supervised Monocular Depth Estimation for Dynamic Scenes](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
405. Y. Liu, J. Zhang, D. Peng, M. Huang, X. Wang, J. Tang, C. Huang, D. Lin, C. Shen, X. Bai, L. Jin (2023), “[SPTS v2: Single-Point Scene Text Spotting](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
404. J. Liu, B. Zhuang, P. Chen, C. Shen, J. Cai, M. Tan (2023), “[Single-Path Bit Sharing for Automatic Loss-Aware Model Compression](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
403. Y. Xie, J. Zhang, Y. Xia, C. Shen (2023), “[Learning From Partially Labeled Data for Multi-Organ and Tumor Segmentation](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
402. M. Lin, M. Chen, Y. Zhang, C. Shen, R. Ji, L. Cao (2023), “[Super Vision Transformer](#)”, *Int. J. Computer Vision*.

401. H. Xiong, H. Lu, C. Liu, L. Liu, C. Shen, Z. Cao (2023), “[From Open Set to Closed Set: Supervised Spatial Divide-and-Conquer for Object Counting](#)”, *Int. J. Computer Vision*.
400. Y. Yan, Y. Shu, S. Chen, J. Xue, C. Shen, H. Wang (2023), “[SPL-Net: Spatial-Semantic Patch Learning Network for Facial Attribute Recognition With Limited Labeled Data](#)”, *Int. J. Computer Vision*.
399. B. Zhang, L. Liu, M.H. Phan, Z. Tian, C. Shen, Y. Liu (2023), “[SegViT V2: Exploring Efficient and Continual Semantic Segmentation With Plain Vision Transformers](#)”, *Int. J. Computer Vision*.
398. Y. Xi, H. Chen, N. Wang, P. Wang, Y. Zhang, C. Shen, Y. Liu (2023), “[A Dynamic Feature Interaction Framework for Multi-Task Visual Perception](#)”, *Int. J. Computer Vision*.
397. N. Sai, J.P. Bockman, H. Chen, N. Watson-Haigh, B. Xu, X. Feng, A. Piechatzek, C. Shen, M. Gilliam (2023), “[SAI: An Efficient and User-Friendly Tool for Measurement of Stomatal Pores and Density Using Deep Computer Vision](#)”, *New Phytologist*.
396. X. Wang, Z. Yu, S. De Mello, J. Kautz, A. Anandkumar, C. Shen, J. Alvarez (2022), “[FreeSOLO: Learning to Segment Objects Without Annotations](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
395. A. Long, W. Yin, T. Ajanthan, V. Nguyen, P. Purkait, R. Garg, A. Blair, C. Shen, A. van den Hengel (2022), “[Retrieval Augmented Classification for Long-Tail Visual Recognition](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
394. Y. Dai, B. Price, H. Zhang, C. Shen (2022), “[Boosting Robustness of Image Matting With Context Assembling and Strong Data Augmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
393. W. Zhang, Z. Huang, G. Yu, T. Chen, G. Luo, X. Wang, W. Liu, C. Shen (2022), “[TopFormer: Token Pyramid Transformer for Mobile Semantic Segmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
392. C. Ding, G. Pan, C. Shen (2022), “[Catching Both Gray and Black Swans: Open-Set Supervised Anomaly Detection](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
391. R. Li, C. Zhang, G. Lin, Z. Wang, C. Shen (2022), “[RigidFlow: Self-Supervised Scene Flow Learning on Point Clouds by Local Rigidity Prior](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'22)*.
390. Y. Gao, J. Zhuang, S. Lin, H. Cheng, X. Sun, K. Li, C. Shen (2022), “[DisCo: Remedying Self-Supervised Learning on Lightweight Models With Distilled Contrastive Learning](#)” [Oral presentation], *European Conf. Computer Vision (ECCV'22)*.
389. P. Chen, M. Zhang, Y. Shen, K. Sheng, Y. Gao, X. Sun, K. Li, C. Shen (2022), “[Efficient Decoder-Free Object Detection With Transformers](#)”, *European Conf. Computer Vision (ECCV'22)*.
388. T. He, W. Yin, C. Shen, A. van den Hengel (2022), “[PointInst3D: Segmenting 3D Instances by Points](#)”, *European Conf. Computer Vision (ECCV'22)*.
387. W. Mao, Y. Ge, C. Shen, Z. Tian, X. Wang, Z. Wang, A. van den Hengel (2022), “[Poseur: Direct Human Pose Regression With Transformers](#)”, *European Conf. Computer Vision (ECCV'22)*.
386. S. Jia, B. Yin, T. Yao, S. Ding, C. Shen, X. Yang, C. Ma (2022), “[Adv-Attribute: Inconspicuous and Transferable Adversarial Attack on Face Recognition](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
385. B. Zhang, Z. Tian, Q. Tang, X. Chu, X. Wei, C. Shen, Y. Liu (2022), “[SegViT: Semantic Segmentation With Plain Vision Transformers](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
384. C. Lin, A. Wu, J. Liang, J. Zhang, W. Ge, W. Zheng, C. Shen (2022), “[Text-Adaptive Multiple Visual Prototype Matching for Video-Text Retrieval](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
383. Z. Tian, X. Chu, X. Wang, X. Wei, C. Shen (2022), “[Fully Convolutional One-Stage 3D Object Detection on LiDAR Range Images](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.

382. C. Zhang, W. Yin, Z. Wang, G. Yu, B. Fu, C. Shen (2022), “[Hierarchical Normalization for Robust Monocular Depth Estimation](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
381. J. Liang, E. Zhang, J. Zhang, C. Shen (2022), “[Multi-Dataset Training of Transformers for Robust Action Recognition](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
380. Y. Gao, J. Liu, Z. Xu, J. Zhang, K. Li, R. Ji, C. Shen (2022), “[PyramidCLIP: Hierarchical Feature Alignment for Vision-Language Model Pretraining](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
379. J. Zhang, C. Chen, B. Li, L. Lyu, S. Wu, S. Ding, C. Shen, C. Wu (2022), “[DENSE: Data-Free One-Shot Federated Learning](#)”, *Advances in Neural Information Processing Systems (NeurIPS'22)*.
378. D. Peng, X. Wang, Y. Liu, J. Zhang, M. Huang, S. Lai, S. Zhu, J. Li, D. Lin, C. Shen, X. Bai, L. Jin (2022), “[SPTS: Single-Point Text Spotting](#)”, *ACM Int. Conf. Multimedia (ACMMM'22)*.
377. Z. Tian, C. Shen, H. Chen, T. He (2022), “[FCOS: A Simple and Strong Anchor-Free Object Detector](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
376. T. He, C. Shen, A. van den Hengel (2022), “[Dynamic Convolution for 3D Point Cloud Instance Segmentation](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
375. C. Zhang, Y. Cai, G. Lin, C. Shen (2022), “[DeepEMD: Differentiable Earth Mover'S Distance for Few-Shot Learning](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
374. W. Yin, J. Zhang, O. Wang, S. Niklaus, S. Chen, Y. Liu, C. Shen (2022), “[Towards Accurate Reconstruction of 3D Scene Shape From a Single Monocular Image](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
373. Z. Tian, B. Zhang, H. Chen, C. Shen (2022), “[Instance and Panoptic Segmentation Using Conditional Convolutions](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
372. B. Zhuang, C. Shen, M. Tan, P. Chen, L. Liu, I. Reid (2022), “[Structured Binary Neural Networks for Image Recognition](#)”, *Int. J. Computer Vision*.
371. L. Wang, H. Zhang, Q. Xiao, H. Xu, C. Shen, X. Jin (2022), “[Effective Eyebrow Matting With Domain Adaptation](#)”, *Computer Graphics Forum*.
370. X. Wang, R. Zhang, C. Shen, T. Kong (2022), “[DenseCL: A Simple Framework for Self-Supervised Dense Visual Pre-Training](#)”, *Visual Informatics*.
369. L. Cheng, P. Fang, Y. Liang, L. Zhang, C. Shen, H. Wang (2022), “[TSGB: Target-Selective Gradient Backprop for Probing CNN Visual Saliency](#)”, *IEEE Trans. Image Processing*.
368. L. Sun, W. Yin, E. Xie, Z. Li, C. Sun, C. Shen (2022), “[Improving Monocular Visual Odometry Using Learned Depth](#)”, *IEEE Trans. Robotics*.
367. Y. Cai, Y. Liu, ChunhuaShen, L. Jin, Y. Li, D. Ergu (2022), “[Arbitrarily Shaped Scene Text Detection With Dynamic Convolution](#)”, *Pattern Recognition*.
366. J. Zhang, Y. Xie, Y. Xia, C. Shen (2021), “[DoDNet: Learning to Segment Multi-Organ and Tumors From Multiple Partially Labeled Datasets](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
365. Z. Tian, C. Shen, X. Wang, H. Chen (2021), “[BoxInst: High-Performance Instance Segmentation With Box Annotations](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
364. P. Chen, J. Liu, B. Zhuang, M. Tan, C. Shen (2021), “[AQD: Towards Accurate Quantized Object Detection](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
363. W. Mao, Z. Tian, X. Wang, C. Shen (2021), “[FCPose: Fully Convolutional Multi-Person Pose Estimation With Dynamic Instance-Aware Convolutions](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
362. Y. Liu, W. Yin, Y. Chen, H. Chen, C. Shen (2021), “[Generic Perceptual Loss for Modelling Structured Output Dependencies](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.

361. W. Yin, J. Zhang, O. Wang, S. Niklaus, L. Mai, S. Chen, C. Shen (2021), “[Learning to Recover 3D Scene Shape From a Single Image](#)” [Listed as one of the Best Paper Candidates, 32 out of about 6000 submissions], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
360. T. He, C. Shen, A. van den Hengel (2021), “[DyCo3D: Robust Instance Segmentation of 3D Point Clouds Through Dynamic Convolution](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
359. Y. Dai, H. Lu, C. Shen (2021), “[Learning Affinity-Aware Upsampling for Deep Image Matting](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
358. D. Ruan, Y. Yan, S. Lai, Z. Chai, C. Shen, H. Wang (2021), “[Feature Decomposition and Reconstruction Learning for Effective Facial Expression Recognition](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
357. Y. Shu, Y. Yan, S. Chen, J. Xue, C. Shen, H. Wang (2021), “[Learning Spatial-Semantic Relationship for Facial Attribute Recognition With Limited Labeled Data](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
356. Y. Wang, Z. Xu, X. Wang, C. Shen, B. Cheng, H. Shen, H. Xia (2021), “[End-to-End Video Instance Segmentation With Transformers](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
355. R. Li, G. Lin, T. He, F. Liu, C. Shen (2021), “[HCRF-Flow: Scene Flow From Point Clouds With Continuous High-Order CRFs and Position-Aware Flow Embedding](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
354. X. Wang, R. Zhang, C. Shen, T. Kong, L. Li (2021), “[Dense Contrastive Learning for Self-Supervised Visual Pre-Training](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
353. D. Guo, Y. Shao, Y. Cui, Z. Wang, L. Zhang, C. Shen (2021), “[Graph Attention Tracking](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'21)*.
352. P. Chen, B. Zhuang, C. Shen (2021), “[FATNN: Fast and Accurate Ternary Neural Networks](#)”, *IEEE Int. Conf. Computer Vision (ICCV'21)*.
351. C. Yan, G. Pang, J. Jiao, X. Bai, X. Feng, C. Shen (2021), “[Occluded Person Re-Identification With Single-Scale Global Representations](#)” [Oral presentation], *IEEE Int. Conf. Computer Vision (ICCV'21)*.
350. C. Yan, G. Pang, L. Wang, J. Jiao, X. Feng, C. Shen, J. Li (2021), “[BV-Person: A Large-Scale Dataset for Bird-View Person Re-Identification](#)”, *IEEE Int. Conf. Computer Vision (ICCV'21)*.
349. J. Yuan, Y. Liu, C. Shen, Z. Wang, H. Li (2021), “[A Simple Baseline for Semi-Supervised Semantic Segmentation With Strong Data Augmentation](#)”, *IEEE Int. Conf. Computer Vision (ICCV'21)*.
348. C. Shu, Y. Liu, J. Gao, L. Xu, C. Shen (2021), “[Channel-Wise Knowledge Distillation for Dense Prediction](#)”, *IEEE Int. Conf. Computer Vision (ICCV'21)*.
347. C. Zhang, H. Ding, G. Lin, R. Li, C. Wang, C. Shen (2021), “[Meta Navigator: Search for a Good Adaptation Policy for Few-Shot Learning](#)”, *IEEE Int. Conf. Computer Vision (ICCV'21)*.
346. X. Chu, Z. Tian, Y. Wang, B. Zhang, H. Ren, X. Wei, H. Xia, C. Shen (2021), “[Twins: Revisiting the Design of Spatial Attention in Vision Transformers](#)”, *Advances in Neural Information Processing Systems (NeurIPS'21)*.
345. B. Zhang, Y. Liu, Z. Tian, C. Shen (2021), “[Dynamic Neural Representational Decoders for High-Resolution Semantic Segmentation](#)”, *Advances in Neural Information Processing Systems (NeurIPS'21)*.
344. G. Pang, A. van den Hengel, C. Shen, L. Cao (2021), “[Toward Deep Supervised Anomaly Detection: Reinforcement Learning From Partially Labeled Anomaly Data](#)”, *ACM SIGKDD Conf. Knowledge Discovery & Data Mining (KDD'21)*.
343. X. Zhang, X. Wang, J. Bian, C. Shen, M. You (2021), “[Diverse Knowledge Distillation for End-to-End Person Search](#)”, *AAAI Conf. Artificial Intelligence (AAAI'21)*.

342. C. Liu, P. Chen, B. Zhuang, C. Shen, B. Zhang, W. Ding (2021), “SA-BNN: State-Aware Binary Neural Network”, *AAAI Conf. Artificial Intelligence (AAAI'21)*.
341. Y. Xie, J. Zhang, C. Shen, Y. Xia (2021), “CoTr: Efficient 3D Medical Image Segmentation by Bridging CNN and Transformer”, *Int. Conf. Medical Image Computing & Computer Assisted Intervention (MICCAI'21)*.
340. L. Kong, C. Shen, J. Yang (2021), “FastFlowNet: A Lightweight Network for Fast Optical Flow Estimation”, *Int. Conf. Robotics & Automation (ICRA'21)*.
339. H. Wang, P. Chen, B. Zhuang, C. Shen (2021), “Fully Quantized Image Super-Resolution Networks”, *ACM Int. Conf. Multimedia (ACMMM'21)*.
338. Y. Zhuge, C. Shen (2021), “Deep Reasoning Network for Few-Shot Semantic Segmentation”, *ACM Int. Conf. Multimedia (ACMMM'21)*.
337. J. Bian, H. Zhan, N. Wang, T. Chin, C. Shen, I. Reid (2021), “Auto-Rectify Network for Unsupervised Indoor Depth Estimation”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
336. X. Wang, R. Zhang, C. Shen, T. Kong, L. Li (2021), “SOLO: A Simple Framework for Instance Segmentation”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
335. Y. Liu, C. Shen, L. Jin, T. He, P. Chen, C. Liu, H. Chen (2021), “ABCNet V2: Adaptive Bezier-Curve Network for Real-Time End-to-End Text Spotting”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
334. W. Yin, Y. Liu, C. Shen (2021), “Virtual Normal: Enforcing Geometric Constraints for Accurate and Robust Depth Prediction”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
333. P. Wang, H. Li, C. Shen (2021), “Towards End-to-End Text Spotting in Natural Scenes”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
332. W. Wang, E. Xie, X. Li, X. Liu, D. Liang, Z. Yang, T. Lu, C. Shen (2021), “PAN++: Towards Efficient and Accurate End-to-End Spotting of Arbitrarily-Shaped Text”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
331. B. Zhuang, J. Liu, M. Tan, L. Liu, I. Reid, C. Shen (2021), “Effective Training of Convolutional Neural Networks With Low-Bitwidth Weights and Activations”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
330. Y. Cui, D. Guo, Y. Shao, Z. Wang, C. Shen, L. Zhang, S. Chen (2021), “Joint Classification and Regression for Visual Tracking With Fully Convolutional Siamese Networks”, *Int. J. Computer Vision*.
329. H. Zhang, Y. Li, H. Chen, C. Gong, Z. Bai, C. Shen (2021), “Memory-Efficient Hierarchical Neural Architecture Search for Image Restoration”, *Int. J. Computer Vision*.
328. Q. Yan, D. Gong, Q. Shi, A. van den Hengel, C. Shen, I. Reid, Y. Zhang (2021), “A Dual-Attention-Guided Network for Ghost-Free High Dynamic Range Imaging”, *Int. J. Computer Vision*.
327. C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, N. Sang (2021), “BiSeNet V2: Bilateral Network With Guided Aggregation for Real-Time Semantic Segmentation”, *Int. J. Computer Vision*.
326. N. Wang, Y. Gao, H. Chen, P. Wang, Z. Tian, C. Shen, Y. Zhang (2021), “NAS-FCOS: Efficient Search for Object Detection Architectures”, *Int. J. Computer Vision*.
325. J. Bian, H. Zhan, N. Wang, Z. Li, L. Zhang, C. Shen, M. Cheng, I. Reid (2021), “Unsupervised Scale-Consistent Depth Learning From Video”, *Int. J. Computer Vision*.
324. Y. Liu, T. He, H. Chen, X. Wang, C. Luo, S. Zhang, C. Shen, L. Jin (2021), “Exploring the Capacity of an Orderless Box Discretization Network for Multi-Orientation Scene Text Detection”, *Int. J. Computer Vision*.
323. Y. Xie, J. Zhang, Z. Liao, J. Verjans, C. Shen, Y. Xia (2021), “Intra- and Inter-Pair Consistency for Semi-Supervised Gland Segmentation”, *IEEE Trans. Image Processing*.
322. Y. Zhao, X. Yu, Y. Gao, C. Shen (2021), “Learning Discriminative Region Representation for Person Retrieval”, *Pattern Recognition*.

321. Y. Zhao, C. Shen, X. Yu, H. Chen, Y. Gao, S. Xiong (2021), “[Learning Deep Part-Aware Embedding for Person Retrieval](#)”, *Pattern Recognition*.
320. L. Tian, P. Wang, G. Liang, C. Shen (2021), “[An Adversarial Human Pose Estimation Network Injected With Graph Structure](#)”, *Pattern Recognition*.
319. Y. Qi, Q. Wu, P. Anderson, X. Wang, W.Y. Wang, C. Shen, A. van den Hengel (2020), “[REVERIE: Remote Embodied Visual Referring Expression in Real Indoor Environments](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
318. C. Yu, J. Wang, C. Gao, G. Yu, C. Shen, N. Sang (2020), “[Context Prior for Scene Segmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
317. R. Zhang, Z. Tian, C. Shen, M. You, Y. Yan (2020), “[Mask Encoding for Single Shot Instance Segmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
316. X. Wang, Y. Liu, C. Shen, C.C. Ng, C. Luo, L. Jin, C.S. Chan, A. van den Hengel, L. Wang (2020), “[On the General Value of Evidence, and Bilingual Scene-Text Visual Question Answering](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
315. B. Zhuang, L. Liu, M. Tan, C. Shen, I. Reid (2020), “[Training Quantized Neural Networks With a Full-Precision Auxiliary Module](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
314. Y. Liu, H. Chen, C. Shen, T. He, L. Jin, L. Wang (2020), “[ABCNet: Arbitrarily-Shaped Scene Text Spotting With Adaptive Bezier-Curve Network in Real Time](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
313. H. Chen, K. Sun, Z. Tian, C. Shen, Y. Huang, Y. Yan (2020), “[BlendMask: Top-Down Meets Bottom-up for Instance Segmentation](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
312. E. Xie, P. Sun, X. Song, W. Wang, X. Liu, D. Liang, C. Shen, P. Luo (2020), “[PolarMask: Single Shot Instance Segmentation With Polar Representation](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
311. C. Zhang, Y. Cai, G. Lin, C. Shen (2020), “[DeepEMD: Few-Shot Image Classification With Differentiable Earth Mover'S Distance and Structured Classifiers](#)” [Oral presentation], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
310. N. Wang, Y. Gao, H. Chen, P. Wang, Z. Tian, C. Shen, Y. Zhang (2020), “[NAS-FCOS: Fast Neural Architecture Search for Object Detection](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
309. H. Zhang, Y. Li, H. Chen, C. Shen (2020), “[Memory-Efficient Hierarchical Neural Architecture Search for Image Denoising](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
308. G. Pang, C. Yan, C. Shen, A. van den Hengel, X. Bai (2020), “[Self-Trained Deep Ordinal Regression for End-to-End Video Anomaly Detection](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'20)*.
307. H. Wang, Q. Wu, C. Shen (2020), “[Soft Expert Reward Learning for Vision-and-Language Navigation](#)”, *European Conf. Computer Vision (ECCV'20)*.
306. W. Wang, E. Xie, X. Liu, W. Wang, D. Liang, C. Shen, X. Bai (2020), “[Scene Text Image Super-Resolution in the Wild](#)”, *European Conf. Computer Vision (ECCV'20)*.
305. W. Wang, X. Liu, X. Ji, E. Xie, D. Liang, Z. Yang, T. Lu, C. Shen, P. Luo (2020), “[AE TextSpotter: Learning Visual and Linguistic Representation for Ambiguous Text Spotting](#)”, *European Conf. Computer Vision (ECCV'20)*.
304. E. Xie, W. Wang, W. Wang, M. Ding, C. Shen, P. Luo (2020), “[Segmenting Transparent Objects in the Wild](#)”, *European Conf. Computer Vision (ECCV'20)*.

303. Z. Tian, C. Shen, H. Chen (2020), “[Conditional Convolutions for Instance Segmentation](#)” [Oral presentation], *European Conf. Computer Vision (ECCV'20)*.
302. X. Wang, T. Kong, C. Shen, Y. Jiang, L. Li (2020), “[SOLO: Segmenting Objects by Locations](#)”, *European Conf. Computer Vision (ECCV'20)*.
301. T. He, D. Gong, Z. Tian, C. Shen (2020), “[Learning and Memorizing Representative Prototypes for 3D Point Cloud Semantic and Instance Segmentation](#)”, *European Conf. Computer Vision (ECCV'20)*.
300. C. Yu, Y. Liu, C. Gao, C. Shen, N. Sang (2020), “[Representative Graph Neural Network](#)”, *European Conf. Computer Vision (ECCV'20)*.
299. L. Liu, H. Lu, H. Zou, H. Xiong, Z. Cao, C. Shen (2020), “[Weighing Counts: Sequential Crowd Counting by Reinforcement Learning](#)”, *European Conf. Computer Vision (ECCV'20)*.
298. Y. Liu, C. Shen, C. Yu, J. Wang (2020), “[Efficient Semantic Video Segmentation With per-Frame Inference](#)”, *European Conf. Computer Vision (ECCV'20)*.
297. T. He, Y. Liu, C. Shen, X. Wang, C. Sun (2020), “[Instance-Aware Embedding for Point Cloud Instance Segmentation](#)”, *European Conf. Computer Vision (ECCV'20)*.
296. X. Wang, R. Zhang, T. Kong, L. Li, C. Shen (2020), “[SOLOv2: Dynamic and Fast Instance Segmentation](#)”, *Advances in Neural Information Processing Systems (NeurIPS'20)*.
295. D. Teney, P. Wang, J. Cao, L. Liu, C. Shen, A. van den Hengel (2020), “[V-PROM: A Benchmark for Visual Reasoning Using Visual Progressive Matrices](#)”, *AAAI Conf. Artificial Intelligence (AAAI'20)*.
294. X. Wang, W. Yin, T. Kong, Y. Jiang, L. Li, C. Shen (2020), “[Task-Aware Monocular Depth Estimation for 3D Object Detection](#)”, *AAAI Conf. Artificial Intelligence (AAAI'20)*.
293. H. Wang, G. Pang, C. Shen, C. Ma (2020), “[Unsupervised Representation Learning by Predicting Random Distances](#)”, *Int. Joint Conferences on Artificial Intelligence (IJCAI'20)*.
292. Y. Xie, J. Zhang, Z. Liao, C. Shen, J. Verjans, Y. Xia (2020), “[Pairwise Relation Learning for Semi-Supervised Gland Segmentation](#)”, *Int. Conf. Medical Image Computing & Computer Assisted Intervention (MICCAI'20)*.
291. J. Cao, Y. Guo, Q. Wu, C. Shen, J. Huang, M. Tan (2020), “[Improving Generative Adversarial Networks With Local Coordinate Coding](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
290. H. Lu, Y. Dai, C. Shen, S. Xu (2020), “[Index Networks](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
289. Y. Liu, C. Shun, J. Wang, C. Shen (2020), “[Structured Knowledge Distillation for Dense Prediction](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
288. Y. Zhou, R. Ji, J. Su, X. Sun, D. Meng, Y. Gao, C. Shen (2020), “[Plenty Is Plague: Fine-Grained Learning for Visual Question Answering](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
287. L. Zhang, Z. Shi, J.T. Zhou, M. Cheng, Y. Liu, J. Bian, Z. Zeng, C. Shen (2020), “[Ordered or Orderless: A Revisit for Video Based Person Re-Identification](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
286. Y. Chen, C. Shen, H. Chen, X. Wei, L. Liu, J. Yang (2020), “[Adversarial Learning of Structure-Aware Fully Convolutional Networks for Landmark Localization](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
285. C. Luo, Q. Lin, Y. Liu, L. Jin, C. Shen (2020), “[Separating Content From Style Using Adversarial Learning for Recognizing Text in the Wild](#)”, *Int. J. Computer Vision*.
284. J. Zhang, Y. Xie, Z. Liao, G. Pang, J. Verjans, W. Li, Z. Sun, J. He, Y. Li, C. Shen, Y. Xia (2020), “[Viral Pneumonia Screening on Chest X-Ray Images Using Confidence-Aware Anomaly Detection](#)”, *IEEE Trans. Medical Imaging*.
283. G. Pang, C. Shen, L. Cao, A. van den Hengel (2020), “[Deep Learning for Anomaly Detection: A Review](#)”, *ACM Computing Surveys*.

282. Y. Dai, H. Lu, C. Shen (2020), “Towards Light-Weight Portrait Matting Via Parameter Sharing”, *Computer Graphics Forum*.
281. Y. Xie, J. Zhang, H. Lu, C. Shen, Y. Xia (2020), “SESV: Accurate Medical Image Segmentation by Predicting and Correcting Errors”, *IEEE Trans. Medical Imaging*.
280. X. Zhang, R. Zhang, J. Cao, D. Gong, M. You, C. Shen (2020), “Part-Guided Attention Learning for Vehicle Instance Retrieval”, *IEEE Trans. Intelligent Transportation Systems*.
279. L. Liu, Z. Cao, H. Lu, H. Xiong, C. Shen (2020), “NSSNet: Scale-Aware Object Counting With Non-Scale Suppression”, *IEEE Trans. Circuits & Systems for Video Technology*.
278. W. Liu, P. Zhang, X. Huang, J. Yang, C. Shen, I. Reid (2020), “Real-Time Image Smoothing Via Iterative Least Squares”, *ACM Trans. Graphics*.
277. L. Zhang, P. Wang, L. Liu, C. Shen, W. Wei, Y. Zhang, A. van den Hengel (2020), “Towards Effective Deep Embedding for Zero-Shot Learning”, *IEEE Trans. Circuits & Systems for Video Technology*.
276. G. Dong, Y. Yan, C. Shen, H. Wang (2020), “Real-Time High-Performance Semantic Image Segmentation of Urban Street Scenes”, *IEEE Trans. Intelligent Transportation Systems*.
275. L. Zhang, P. Wang, H. Li, Z. Li, C. Shen, Y. Zhang (2020), “A Robust Attentional Framework for License Plate Recognition in the Wild”, *IEEE Trans. Intelligent Transportation Systems*.
274. Y. Xie, J. Zhang, Y. Xia, C. Shen (2020), “A Mutual Bootstrapping Model for Automated Skin Lesion Segmentation and Classification”, *IEEE Trans. Medical Imaging*.
273. S. Zhang, Y. Liu, L. Jin, Z. Wei, C. Shen (2020), “OPMP: An Omni-Directional Pyramid Mask Proposal Network for Arbitrary-Shape Scene Text Detection”, *IEEE Trans. Multimedia*.
272. Y. Yan, Y. Huang, S. Chen, C. Shen, H. Wang (2020), “Joint Deep Learning of Facial Expression Synthesis and Recognition”, *IEEE Trans. Multimedia*.
271. X. Peng, H. Zhu, J. Feng, C. Shen, H. Zhang, J. Zhou (2020), “Deep Clustering With Sample-Assignment Invariance Prior”, *IEEE Trans. Neural Networks & Learning Systems*.
270. D. Gong, Z. Zhang, Q. Shi, A. van den Hengel, C. Shen, Y. Zhang (2020), “Learning Deep Gradient Descent Optimization for Image Deconvolution”, *IEEE Trans. Neural Networks & Learning Systems*.
269. H. Xiong, Z. Cao, H. Lu, S. Madec, L. Liu, C. Shen (2020), “TasselNetv2: In-Field Counting of Wheat Spikes With Context-Augmented Local Regression Networks”, *Plant Methods*.
268. Y. Zhao, Y. Liu, C. Shen, Y. Gao, S. Xiong (2020), “MobileFAN: Transferring Deep Hidden Representation for Face Alignment”, *Pattern Recognition*.
267. L. Zhang, W. Wei, Q. Shi, C. Shen, A. van den Hengel, Y. Zhang (2020), “Accurate Tensor Completion Via Adaptive Low-Rank Representation”, *IEEE Trans. Neural Networks & Learning Systems*.
266. P. Wang, Q. Wu, J. Cao, C. Shen, L. Gao, A. vanden Hengel (2019), “Neighbourhood Watch: Referring Expression Comprehension Via Language-Guided Graph Attention Networks”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
265. V. Nekrasov, H. Chen, C. Shen, I. Reid (2019), “Fast Neural Architecture Search of Compact Semantic Segmentation Models Via Auxiliary Cells”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
264. H. Li, P. Wang, C. Shen, A. van den Hengel (2019), “Visual Question Answering as Reading Comprehension”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
263. B. Zhuang, C. Shen, M. Tan, L. Liu, I. Reid (2019), “Structured Binary Neural Networks for Accurate Image Classification and Semantic Segmentation”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
262. X. Wang, S. Liu, X. Shen, C. Shen, J. Jia (2019), “Associatively Segmenting Instances and Semantics in Point Clouds”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.

261. C. Zhang, G. Lin, F. Liu, R. Yao, C. Shen (2019), “[CANet: Class-Agnostic Segmentation Networks With Iterative Refinement and Attentive Few-Shot Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
260. J. Zhang, Q. Wu, J. Zhang, C. Shen, J. Lu (2019), “[Mind Your Neighbours: Image Annotation With Metadata Neighbourhood Graph Co-Attention Networks](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
259. Z. Tian, T. He, C. Shen, Y. Yan (2019), “[Decoders Matter for Semantic Segmentation: Data-Dependent Decoding Enables Flexible Feature Aggregation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
258. T. He, C. Shen, Z. Tian, D. Gong, C. Sun, Y. Yan (2019), “[Knowledge Adaptation for Efficient Semantic Segmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
257. Q. Yan, D. Gong, Q. Shi, A. van den Hengel, C. Shen, I. Reid, Y. Zhang (2019), “[Attention-Guided Network for Ghost-Free High Dynamic Range Imaging](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'19)*.
256. W. Wang, E. Xie, X. Song, Y. Zang, W. Wang, T. Lu, G. Yu, C. Shen (2019), “[Efficient and Accurate Arbitrary-Shaped Text Detection With Pixel Aggregation Network](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
255. H. Lu, Y. Dai, C. Shen, S. Xu (2019), “[Indices Matter: Learning to Index for Deep Image Matting](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
254. X. Zhang, J. Cao, C. Shen, M. You (2019), “[Self-Training With Progressive Augmentation for Unsupervised Cross-Domain Person Re-Identification](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
253. W. Yin, Y. Liu, C. Shen, Y. Yan (2019), “[Enforcing Geometric Constraints of Virtual Normal for Depth Prediction](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
252. H. Zhang, C. Shen, Y. Li, Y. Cao, Y. Liu, Y. Yan (2019), “[Exploiting Temporal Consistency for Real-Time Video Depth Estimation](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
251. Z. Tian, C. Shen, H. Chen, T. He (2019), “[FCOS: Fully Convolutional One-Stage Object Detection](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
250. H. Xiong, H. Lu, C. Liu, L. Liu, Z. Cao, C. Shen (2019), “[From Open Set to Closed Set: Counting Objects by Spatial Divide-and-Conquer](#)”, *IEEE Int. Conf. Computer Vision (ICCV'19)*.
249. J. Bian, Z. Li, N. Wang, H. Zhan, C. Shen, M. Cheng, I. Reid (2019), “[Unsupervised Scale-Consistent Depth and Ego-Motion Learning From Monocular Video](#)”, *Advances in Neural Information Processing Systems (NeurIPS'19)*.
248. J. Cao, L. Mo, Y. Zhang, K. Jia, C. Shen, M. Tan (2019), “[Multi-Marginal Wasserstein GAN](#)”, *Advances in Neural Information Processing Systems (NeurIPS'19)*.
247. G. Pang, C. Shen, A. van den Hengel (2019), “[Deep Anomaly Detection With Deviation Networks](#)”, *ACM SIGKDD Conf. Knowledge Discovery & Data Mining (KDD'19)*.
246. H. Li, P. Wang, C. Shen, G. Zhang (2019), “[Show, Attend and Read: A Simple and Strong Baseline for Irregular Text Recognition](#)”, *AAAI Conf. Artificial Intelligence (AAAI'19)*.
245. J. Zhang, Y. Xie, P. Zhang, H. Chen, Y. Xia, C. Shen (2019), “[Light-Weight Hybrid Convolutional Network for Liver Tumor Segmentation](#)”, *Int. Joint Conf. Artificial Intelligence (IJCAI'19)*.
244. C. Yan, G. Pang, X. Bai, C. Shen, J. Zhou, E. Hancock (2019), “[Deep Hashing by Discriminating Hard Examples](#)”, *ACM Int. Conf. Multimedia (ACMMM'19)*.
243. Y. Xie, H. Lu, J. Zhang, C. Shen, Y. Xia (2019), “[Deep Segmentation-Emendation Model for Gland Instance Segmentation](#)”, *Int. Conf. Medical Image Computing & Computer Assisted Intervention (MICCAI'19)*.

242. V. Nekrasov, T. Dharmasiri, A. Spek, T. Drummond, C. Shen, I. Reid (2019), “[Real-Time Joint Semantic Segmentation and Depth Estimation Using Asymmetric Annotations](#)”, *Int. Conf. Robotics & Automation (ICRA'19)*.
241. G. Lin, F. Liu, A. Milan, C. Shen, I. Reid (2019), “[RefineNet: Multi-Path Refinement Networks for Dense Prediction](#)” [[Pytorch code is \[https://github.com/DrSleep/refinenet-pytorch here\]](#)], *IEEE Trans. Pattern Analysis & Machine Intelligence*.
240. L. Zhang, P. Wang, C. Shen, L. Liu, W. Wei, Y. Zhang, A. van den Hengel (2019), “[Adaptive Importance Learning for Improving Lightweight Image Super-Resolution Network](#)”, *Int. J. Computer Vision*.
239. X. Wei, H. Ye, X. Mu, J. Wu, C. Shen, Z. Zhou (2019), “[Multiple Instance Learning With Emerging Novel Class](#)”, *IEEE Trans. Knowledge & Data Engineering*.
238. T. Zhang, G. Lin, J. Cai, T. Shen, C. Shen, A.C. Kot (2019), “[Decoupled Spatial Neural Attention for Weakly Supervised Semantic Segmentation](#)”, *IEEE Trans. Multimedia*.
237. X. Wang, C. Shen, H. Li, S. Xu (2019), “[Human Detection Aided by Deeply Learned Semantic Masks](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
236. X. Wei, P. Wang, L. Liu, C. Shen, J. Wu (2019), “[Piecewise Classifier Mappings: Learning Fine-Grained Learners for Novel Categories With Few Examples](#)”, *IEEE Trans. Image Processing*.
235. J. Zhang, Q. Wu, J. Zhang, C. Shen, J. Lu, Q. Wu (2019), “[Heritage Image Annotation Via Collective Knowledge](#)”, *Pattern Recognition*.
234. L. Liu, H. Lu, H. Xiong, K. Xian, Z. Cao, C. Shen (2019), “[Counting Objects by Blockwise Classification](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
233. H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, C. Shen (2019), “[Hyperspectral Classification Based on Lightweight 3D-CNN With Transfer Learning](#)”, *IEEE Trans. Geoscience & Remote Sensing*.
232. L. Zhang, W. Wei, Q. Shen, C. Shen, A. van den Hengel (2019), “[Accurate Imagery Recovery Using a Multi-Observation Patch Model](#)”, *Information Sciences*.
231. P. Zhang, W. Liu, H. Lu, C. Shen (2019), “[Salient Object Detection With Lossless Feature Reflection and Weighted Structural Loss](#)”, *IEEE Trans. Image Processing*.
230. P. Wang, L. Liu, C. Shen, H.T. Shen (2019), “[Order-Aware Convolutional Pooling for Video Based Action Recognition](#)”, *Pattern Recognition*.
229. J. Zhang, Y. Xie, Y. Xia, C. Shen (2019), “[Attention Residual Learning for Skin Lesion Classification](#)”, *IEEE Trans. Medical Imaging*.
228. Z. Wu, C. Shen, A. van den Hengel (2019), “[Wider or Deeper: Revisiting the ResNet Model for Visual Recognition](#)”, *Pattern Recognition*.
227. W. Liu, P. Zhang, X. Chen, C. Shen, X. Huang, J. Yang (2019), “[Embedding Bilateral Filter in Least Squares for Efficient Edge-Preserving Image Smoothing](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
226. Y. Zhao, C. Shen, H. Wang, S. Chen (2019), “[Structural Analysis of Attributes for Vehicle Re-Identification and Retrieval](#)”, *IEEE Trans. Intelligent Transportation Systems*.
225. T. Shen, G. Lin, C. Shen, I. Reid (2018), “[Bootstrapping the Performance of Webly Supervised Semantic Segmentation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
224. K. Xian, C. Shen, Z. Cao, H. Lu, Y. Xiao, R. Li, Z. Luo (2018), “[Monocular Relative Depth Perception With Web Stereo Data Supervision](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
223. Y. Song, C. Ma, X. Wu, L. Gong, L. Bao, W. Zuo, C. Shen, R. Lau, M. Yang (2018), “[VITAL: Visual Tracking Via Adversarial Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
222. T. He, Z. Tian, W. Huang, C. Shen, Y. Qiao, C. Sun (2018), “[An End-to-End TextSpotter With Explicit Alignment and Attention](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.

221. Q. Wu, P. Wang, C. Shen, I. Reid, A. van den Hengel (2018), “Are You Talking to Me? Reasoned Visual Dialog Generation Through Adversarial Learning”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
220. B. Zhuang, Q. Wu, C. Shen, I. Reid, A. van den Hengel (2018), “Parallel Attention: A Unified Framework for Visual Object Discovery Through Dialogs and Queries”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
219. B. Zhuang, C. Shen, M. Tan, L. Liu, I. Reid (2018), “Towards Effective Low-Bitwidth Convolutional Neural Networks”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
218. C. Ma, C. Shen, A. Dick, Q. Wu, P. Wang, A. van den Hengel, I. Reid (2018), “Visual Question Answering With Memory-Augmented Networks”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
217. X. Wang, T. Xiao, Y. Jiang, S. Shao, J. Sun, C. Shen (2018), “Repulsion Loss: Detecting Pedestrians in a Crowd” [Others have implemented our paper: [https://github.com/bailvwangzi/repulsion_loss_ssd Repulsion loss in SSD] and [https://github.com/rainofmine/Repulsion_Loss Repulsion loss in RetinaNet]], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
216. Y. Chen, Y. Tai, X. Liu, C. Shen, J. Yang (2018), “FSRNet: End-to-End Learning Face Super-Resolution With Facial Priors”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'18)*.
215. J. Zhang, Q. Wu, C. Shen, J. Zhang, J. Lu, A. van den Hengel (2018), “Goal-Oriented Visual Question Generation Via Intermediate Rewards”, *European Conf. Computer Vision (ECCV'18)*.
214. R. Deng, C. Shen, S. Liu, H. Wang, X. Liu (2018), “Learning to Predict Crisp Boundaries”, *European Conf. Computer Vision (ECCV'18)*.
213. J. Cao, Y. Guo, Q. Wu, C. Shen, J. Huang, M. Tan (2018), “Adversarial Learning With Local Coordinate Coding”, *Int. Conf. Machine Learning (ICML'18)*.
212. J. Zhang, Q. Wu, J. Zhang, C. Shen, J. Lu (2018), “Kill Two Birds With One Stone: Weakly-Supervised Neural Network for Image Annotation and Tag Refinement”, *AAAI Conf. Artificial Intelligence (AAAI'18)*.
211. B. Zhuang, Q. Wu, C. Shen, I. Reid, A. van den Hengel (2018), “HCVRD: A Benchmark for Large-Scale Human-Centered Visual Relationship Detection”, *AAAI Conf. Artificial Intelligence (AAAI'18)*.
210. P. Zhang, W. Liu, H. Lu, C. Shen (2018), “Salient Object Detection by Lossless Feature Reflection”, *Int. Joint Conf. Artificial Intelligence (IJCAI'18)*.
209. L. Dong, Y. Gan, X. Mao, Y. Yang, C. Shen (2018), “Learning Deep Representations Using Convolutional Auto-Encoders With Symmetric Skip Connections”, *IEEE Int. Conf. Acoustics, Speech & Signal Processing (ICASSP'18)*.
208. R. Li, K. Xian, C. Shen, Z. Cao, H. Lu, L. Hang (2018), “Deep Attention-Based Classification Network for Robust Depth Prediction”, *Asian Conf. Computer Vision (ACCV'18)*.
207. X. Wei, C. Zhang, L. Liu, C. Shen, J. Wu (2018), “Coarse-to-Fine: A RNN-Based Hierarchical Attention Model for Vehicle Re-Identification”, *Asian Conf. Computer Vision (ACCV'18)*.
206. V. Nekrasov, C. Shen, I. Reid (2018), “Light-Weight RefineNet for Real-Time Semantic Segmentation”, *British Machine Vision Conference (BMVC'18)*.
205. M. Cai, C. Shen, I. Reid (2018), “A Hybrid Probabilistic Model for Camera Relocalization”, *British Machine Vision Conference (BMVC'18)*.
204. H. Liu, R. Ji, J. Wang, C. Shen (2018), “Ordinal Constraint Binary Coding for Approximate Nearest Neighbor Search”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
203. P. Wang, Q. Wu, C. Shen, A. Dick, A. van den Hengel (2018), “FVQA: Fact-Based Visual Question Answering”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
202. L. Zhang, W. Wei, Y. Zhang, C. Shen, A. van den Hengel, Q. Shi (2018), “Cluster Sparsity Field: An Internal Hyperspectral Imagery Prior for Reconstruction”, *Int. J. Computer Vision*.

201. X. Wei, C. Zhang, J. Wu, C. Shen, Z. Zhou (2018), “[Unsupervised Object Discovery and Co-Localization by Deep Descriptor Transforming](#)”, *Pattern Recognition*.
200. R. Yao, G. Lin, C. Shen, Y. Zhang, Q. Shi (2018), “[Semantics-Aware Visual Object Tracking](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
199. L. Zhang, P. Wang, W. Wei, H. Lu, C. Shen, A. van den Hengel, Y. Zhang (2018), “[Unsupervised Domain Adaptation Using Robust Class-Wise Matching](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
198. H. Li, P. Wang, C. Shen (2018), “[Towards End-to-End Car License Plates Detection and Recognition With Deep Neural Networks](#)”, *IEEE Trans. Intelligent Transportation Systems*.
197. N. Zhuang, Y. Yan, S. Chen, H. Wang, C. Shen (2018), “[Multi-Label Learning Based Deep Transfer Neural Network for Facial Attribute Classification](#)”, *Pattern Recognition*.
196. H. Lu, C. Shen, Z. Cao, Y. Xiao, A. van den Hengel (2018), “[An Embarrassingly Simple Approach to Visual Domain Adaptation](#)”, *IEEE Trans. Image Processing*.
195. M. You, Y. Zhang, C. Shen, X. Zhang (2018), “[An Extended Filtered Channel Framework for Pedestrian Detection](#)”, *IEEE Trans. Intelligent Transportation Systems*.
194. H. Li, P. Wang, M. You, C. Shen (2018), “[Reading Car License Plates Using Deep Neural Networks](#)”, *Image & Vision Computing*.
193. J. Zhang, Q. Wu, C. Shen, J. Zhang, J. Lu (2018), “[Multi-Label Image Classification With Regional Latent Semantic Dependencies](#)”, *IEEE Trans. Multimedia*.
192. G. Guo, H. Wang, C. Shen, Y. Yan, H. Liao (2018), “[Automatic Image Cropping for Visual Aesthetic Enhancement Using Deep Neural Networks and Cascaded Regression](#)”, *IEEE Trans. Multimedia*.
191. Q. Hu, P. Wang, C. Shen, A. van den Hengel, F. Porikli (2018), “[Pushing the Limits of Deep CNNs for Pedestrian Detection](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
190. P. Wang, Q. Wu, C. Shen, A. van den Hengel (2017), “[The VQA-Machine: Learning How to Use Existing Vision Algorithms to Answer New Questions](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
189. G. Lin, A. Milan, C. Shen, I. Reid (2017), “[RefineNet: Multi-Path Refinement Networks for High-Resolution Semantic Segmentation](#)” [<https://github.com/DrSleep/light-weight-refinenet> Light-weight RefineNet with Pytorch code], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
188. Y. Li, G. Lin, B. Zhuang, L. Liu, C. Shen, A. van den Hengel (2017), “[Sequential Person Recognition in Photo Albums With a Recurrent Network](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
187. D. Gong, J. Yang, L. Liu, Y. Zhang, I. Reid, C. Shen, A. van den Hengel, Q. Shi (2017), “[From Motion Blur to Motion Flow: A Deep Learning Solution for Removing Heterogeneous Motion Blur](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
186. B. Zhuang, L. Liu, Y. Li, C. Shen, I. Reid (2017), “[Attend in Groups: A Weakly-Supervised Deep Learning Framework for Learning From Web Data](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
185. P. Wang, L. Liu, C. Shen, Z. Huang, A. van den Hengel, H.T. Shen (2017), “[Multi-Attention Network for One Shot Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'17)*.
184. H. Lu, L. Zhang, Z. Cao, W. Wei, K. Xian, C. Shen, A. van den Hengel (2017), “[When Unsupervised Domain Adaptation Meets Tensor Representations](#)”, *IEEE Int. Conf. Computer Vision (ICCV'17)*.
183. B. Zhuang, L. Liu, C. Shen, I. Reid (2017), “[Towards Context-Aware Interaction Recognition](#)”, *IEEE Int. Conf. Computer Vision (ICCV'17)*.
182. Y. Chen, C. Shen, X. Wei, L. Liu, J. Yang (2017), “[Adversarial PoseNet: A Structure-Aware Convolutional Network for Human Pose Estimation](#)”, *IEEE Int. Conf. Computer Vision (ICCV'17)*.

181. W. Liu, X. Chen, C. Shen, Z. Liu, J. Yang (2017), “[Semi-Global Weighted Least Squares in Image Filtering](#)”, *IEEE Int. Conf. Computer Vision (ICCV'17)*.
180. H. Li, P. Wang, C. Shen (2017), “[Towards End-to-End Text Spotting With Convolutional Recurrent Neural Networks](#)”, *IEEE Int. Conf. Computer Vision (ICCV'17)*.
179. T. Shen, G. Lin, C. Shen, I. Reid (2017), “[Learning Multi-Level Region Consistency With Dense Multi-Label Networks for Semantic Segmentation](#)”, *Int. Joint Conf. Artificial Intelligence (IJCAI'17)*.
178. X. Wei, C. Zhang, Y. Li, C. Xie, J. Wu, C. Shen, Z. Zhou (2017), “[Deep Descriptor Transforming for Image Co-Localization](#)”, *Int. Joint Conf. Artificial Intelligence (IJCAI'17)*.
177. P. Wang, Q. Wu, C. Shen, A. van den Hengel, A. Dick (2017), “[Explicit Knowledge-Based Reasoning for Visual Question Answering](#)”, *Int. Joint Conf. Artificial Intelligence (IJCAI'17)*.
176. S. McMahon, T. Shen, N. Sunderhauf, I. Reid, C. Shen, M. Milford (2017), “[Auxiliary Tasks to Improve Trip Hazard Affordance Detection](#)”, *Australasian Conf. Robotics & Automation (ACRA'17)*.
175. Z. Chen, A. Jacobson, N. Sunderhauf, B. Upcroft, L. Liu, C. Shen, I. Reid, M. Milford (2017), “[Deep Learning Features at Scale for Visual Place Recognition](#)”, *IEEE Int. Conf. Robotics & Automation (ICRA'17)*.
174. R. Qiao, L. Liu, C. Shen, A. van den Hengel (2017), “[Visually Aligned Word Embeddings for Improving Zero-Shot Learning](#)”, *British Machine Vision Conference (BMVC'17)*.
173. T. Shen, G. Lin, L. Liu, C. Shen, I. Reid (2017), “[Weakly Supervised Semantic Segmentation Based on Co-Segmentation](#)”, *British Machine Vision Conference (BMVC'17)*.
172. G. Lin, C. Shen, A. van den Hengel, I. Reid (2017), “[Exploring Context With Deep Structured Models for Semantic Segmentation](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
171. Q. Wu, C. Shen, P. Wang, A. Dick, A. van den Hengel (2017), “[Image Captioning and Visual Question Answering Based on Attributes and External Knowledge](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
170. L. Liu, P. Wang, C. Shen, L. Wang, A. van den Hengel, C. Wang, H.T. Shen (2017), “[Compositional Model Based Fisher Vector Coding for Image Classification](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
169. L. Liu, C. Shen, A. van den Hengel (2017), “[Cross-Convolutional-Layer Pooling for Image Recognition](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
168. G. Lin, F. Liu, C. Shen, J. Wu, H.T. Shen (2017), “[Structured Learning of Binary Codes With Column Generation for Optimizing Ranking Measures](#)”, *Int. J. Computer Vision*.
167. X. Wei, C. Xie, J. Wu, C. Shen (2017), “[Mask-CNN: Localizing Parts and Selecting Descriptors for Bird Species Categorization](#)”, *Pattern Recognition*.
166. H. Lu, Z. Cao, Y. Xiao, B. Zhuang, C. Shen (2017), “[TasselNet: Counting Maize Tassels in the Wild Via Local Counts Regression Network](#)”, *Plant Methods*.
165. Y. Cao, Z. Wu, C. Shen (2017), “[Estimating Depth From Monocular Images as Classification Using Deep Fully Convolutional Residual Networks](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
164. Y. Li, W. Li, C. Shen (2017), “[Removal of Optically Thick Clouds From High-Resolution Satellite Imagery Using Dictionary Group Learning and Interdictionary Nonlocal Joint Sparse Coding](#)”, *IEEE J. Selected Topics in Applied Earth Observations & Remote Sensing*.
163. Q. Hu, H. Wang, T. Li, C. Shen (2017), “[Deep CNNs With Spatially Weighted Pooling for Fine-Grained Car Recognition](#)”, *IEEE Trans. Intelligent Transportation Systems*.
162. F. Liu, G. Lin, R. Qiao, C. Shen (2017), “[Structured Learning of Tree Potentials in CRF for Image Segmentation](#)”, *IEEE Trans. Neural Networks & Learning Systems*.

161. F. Liu, G. Lin, C. Shen (2017), “Discriminative Training of Deep Fully-Connected Continuous CRF With Task-Specific Loss”, *IEEE Trans. Image Processing*.
160. R. Qiao, L. Liu, C. Shen, A. van den Hengel (2017), “Learning Discriminative Trajectorylet Detector Sets for Accurate Skeleton-Based Action Recognition”, *Pattern Recognition*.
159. L. Wu, C. Shen, A. van den Hengel (2017), “Deep Linear Discriminant Analysis on Fisher Networks: A Hybrid Architecture for Person Re-Identification”, *Pattern Recognition*.
158. B. Sheng, C. Shen, G. Lin, J. Li, W. Yang, C. Sun (2017), “Crowd Counting Via Weighted VLAD on Dense Attribute Feature Maps”, *IEEE Trans. Circuits & Systems for Video Technology*.
157. Y. Cao, C. Shen, H.T. Shen (2017), “Exploiting Depth From Single Monocular Images for Object Detection and Semantic Segmentation”, *IEEE Trans. Image Processing*.
156. Q. Wu, D. Teney, P. Wang, C. Shen, A. Dick, A. van den Hengel (2017), “Visual Question Answering: A Survey of Methods and Datasets”, *Computer Vision & Image Understanding*.
155. Q. Wu, P. Wang, C. Shen, A. Dick, A. van den Hengel (2016), “Ask Me Anything: Free-Form Visual Question Answering Based on Knowledge From External Sources”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
154. Q. Wu, C. Shen, L. Liu, A. Dick, A. van den Hengel (2016), “What Value Do Explicit High Level Concepts Have in Vision to Language Problems”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
153. P. Wang, L. Liu, C. Shen, Z. Huang, A. van den Hengel, H.T. Shen (2016), “What'S Wrong With That Object? Identifying Irregular Object From Images by Modelling the Detection Score Distribution”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
152. G. Lin, C. Shen, A. van dan Hengel, I. Reid (2016), “Efficient Piecewise Training of Deep Structured Models for Semantic Segmentation”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
151. B. Zhuang, G. Lin, C. Shen, I. Reid (2016), “Fast Training of Triplet-Based Deep Binary Embedding Networks”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
150. R. Qiao, L. Liu, C. Shen, A. van den Hengel (2016), “Less Is More: Zero-Shot Learning From Online Textual Documents With Noise Suppression”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'16)*.
149. L. Zhang, W. Wei, Y. Zhang, C. Shen, A. van den Hengel, Q. Shi (2016), “Cluster Sparsity Field for Hyperspectral Imagery Denoising”, *European Conf. Computer Vision (ECCV'16)*.
148. Y. Li, L. Liu, C. Shen, A. van den Hengel (2016), “Image Co-Localization by Mimicking a Good Detector'S Confidence Score Distribution”, *European Conf. Computer Vision (ECCV'16)*.
147. X. Mao, C. Shen, Y. Yang (2016), “Image Restoration Using Very Deep Fully Convolutional Encoder-Decoder Networks With Symmetric Skip Connections” [Others have <https://github.com/titu1994/Image-Super-Resolution> implemented our paper]], *Advances in Neural Information Processing Systems (NeurIPS'16)*.
146. P. Wang, C. Shen, A. van den Hengel, P.H.S. Torr (2016), “Large-Scale Binary Quadratic Optimization Using Semidefinite Relaxation and Applications”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
145. F. Liu, C. Shen, G. Lin, I. Reid (2016), “Learning Depth From Single Monocular Images Using Deep Convolutional Neural Fields”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
144. X. Li, C. Shen, A. Dick, Z. Zhang, Y. Zhuang (2016), “Online Metric-Weighted Linear Representations for Robust Visual Tracking”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
143. S. Paisitkriangkrai, C. Shen, A. van den Hengel (2016), “Pedestrian Detection With Spatially Pooled Features and Structured Ensemble Learning”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
142. L. Liu, L. Wang, C. Shen (2016), “A Generalized Probabilistic Framework for Compact Codebook Creation”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.

141. Y. Li, L. Liu, C. Shen, A. van den Hengel (2016), “[Mining Mid-Level Visual Patterns With Deep CNN Activations](#)”, *Int. J. Computer Vision*.
140. P. Wang, C. Shen, A. van den Hengel, P. Torr (2016), “[Efficient Semidefinite Branch-and-Cut for MAP-MRF Inference](#)”, *Int. J. Computer Vision*.
139. C. Zhang, C. Shen, T. Shen (2016), “[Unsupervised Feature Learning for Dense Correspondences Across Scenes](#)”, *Int. J. Computer Vision*.
138. S. Paisitkriangkrai, L. Wu, C. Shen, A. van den Hengel (2016), “[Structured Learning of Metric Ensembles With Application to Person Re-Identification](#)”, *Computer Vision & Image Understanding*.
137. L. Zhang, W. Wei, Y. Zhang, C. Shen, A. van den Hengel, Q. Shi (2016), “[Dictionary Learning for Promoting Structured Sparsity in Hyperspectral Compressive Sensing](#)”, *IEEE Trans. Geoscience & Remote Sensing*.
136. F. Liu, C. Shen, I. Reid, A. van den Hengel (2016), “[Online Unsupervised Feature Learning for Visual Tracking](#)”, *Image & Vision Computing*.
135. P. Wang, Y. Cao, C. Shen, L. Liu, H.T. Shen (2016), “[Temporal Pyramid Pooling Based Convolutional Neural Network for Action Recognition](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
134. H. Li, F. Shen, C. Shen, Y. Yang, Y. Gao (2016), “[Face Recognition Using Linear Representation Ensembles](#)”, *Pattern Recognition*.
133. S. Wang, J. Lu, X. Gu, C. Shen, R. Xia, J. Yang (2016), “[Canonical Principal Angles Correlation Analysis for Two-View Data](#)”, *J. Visual Communication & Image Representation*.
132. R. Yao, Q. Shi, C. Shen, Y. Zhang, A. van den Hengel (2016), “[Part-Based Robust Tracking Using Online Latent Structured Learning](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
131. F. Shen, C. Shen, X. Zhou, Y. Yang, H.T. Shen (2016), “[Face Image Classification by Pooling Raw Features](#)”, *Pattern Recognition*.
130. Q. Hu, S. Paisitkriangkrai, C. Shen, A. van den Hengel, F. Porikli (2016), “[Fast Detection of Multiple Objects in Traffic Scenes With a Common Detection Framework](#)”, *IEEE Trans. Intelligent Transportation Systems*.
129. X. Zhao, X. Li, Z. Zhang, C. Shen, L. Gao, X. Li (2016), “[Scalable Linear Visual Feature Learning Via Online Parallel Nonnegative Matrix Factorization](#)”, *IEEE Trans. Neural Networks & Learning Systems*.
128. Y. Li, L. Liu, C. Shen, A. van den Hengel (2015), “[Mid-Level Deep Pattern Mining](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
127. F. Liu, C. Shen, G. Lin (2015), “[Deep Convolutional Neural Fields for Depth Estimation From a Single Image](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
126. F. Shen, C. Shen, W. Liu, H.T. Shen (2015), “[Supervised Discrete Hashing](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
125. L. Liu, C. Shen, A. van den Hengel (2015), “[The Treasure Beneath Convolutional Layers: Cross Convolutional Layer Pooling for Image Classification](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
124. P. Wang, C. Shen, A. van den Hengel (2015), “[Efficient SDP Inference for Fully-Connected CRFs Based on Low-Rank Decomposition](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
123. S. Paisitkriangkrai, C. Shen, A. van den Hengel (2015), “[Learning to Rank in Person Re-Identification With Metric Ensembles](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
122. M. Tan, Q. Shi, A. van den Hengel, C. Shen, J. Gao, F. Hu, Z. Zhang (2015), “[Learning Graph Structure for Multi-Label Image Classification Via Clique Generation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.

121. B. Li, [C. Shen](#), Y. Dai, A. van den Hengel, M. He (2015), “[Depth and Surface Normal Estimation From Monocular Images Using Regression on Deep Features and Hierarchical CRFs](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'15)*.
120. L. Zhang, W. Wei, Y. Zhang, F. Li, [C. Shen](#), Q. Shi (2015), “[Hyperspectral Compressive Sensing Using Manifold-Structured Sparsity Prior](#)”, *IEEE Int. Conf. Computer Vision (ICCV'15)*.
119. G. Lin, [C. Shen](#), I. Reid, A. van den Hengel (2015), “[Deeply Learning the Messages in Message Passing Inference](#)”, *Advances in Neural Information Processing Systems (NeurIPS'15)*.
118. M. Milford, [C. Shen](#), S. Lowry, N. Suenderhauf, S. Shirazi, G. Lin, F. Liu, E. Pepperell, C. Lerma, B. Upcroft, I. Reid (2015), “[Sequence Searching With Deep-Learnt Depth for Condition- and Viewpoint-Invariant Route-Based Place Recognition](#)” [Best paper award (Sponsored by NVIDIA)], *6th Int. Workshop on Computer Vision in Vehicle Technology, in conjunction with IEEE Conf. Computer Vision & Pattern Recognition (CVVT'15)*.
117. G. Lin, [C. Shen](#), A. van den Hengel (2015), “[Supervised Hashing Using Graph Cuts and Boosted Decision Trees](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
116. M. Harandi, R. Hartley, [C. Shen](#), B. Lovell, C. Sanderson (2015), “[Extrinsic Methods for Coding and Dictionary Learning on Grassmann Manifolds](#)”, *Int. J. Computer Vision*.
115. F. Liu, G. Lin, [C. Shen](#) (2015), “[CRF Learning With CNN Features for Image Segmentation](#)”, *Pattern Recognition*.
114. F. Shen, [C. Shen](#), Q. Shi, A. van den Hengel, Z. Tang, H.T. Shen (2015), “[Hashing on Nonlinear Manifolds](#)”, *IEEE Trans. Image Processing*.
113. H. Li, [C. Shen](#), A. van den Hengel, Q. Shi (2015), “[Worst-Case Linear Discriminant Analysis as Scalable Semidefinite Feasibility Problems](#)”, *IEEE Trans. Image Processing*.
112. L. Luo, [C. Shen](#), X. Liu, C. Zhang (2015), “[A Computational Model of the Short-Cut Rule for 2D Shape Decomposition](#)”, *IEEE Trans. Image Processing*.
111. G. Lin, [C. Shen](#), Q. Shi, A. van den Hengel, D. Suter (2014), “[Fast Supervised Hashing With Decision Trees for High-Dimensional Data](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'14)*.
110. G. Lin, [C. Shen](#), J. Wu (2014), “[Optimizing Ranking Measures for Compact Binary Code Learning](#)”, *European Conf. Computer Vision (ECCV'14)*.
109. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2014), “[Strengthening the Effectiveness of Pedestrian Detection With Spatially Pooled Features](#)”, *European Conf. Computer Vision (ECCV'14)*.
108. L. Liu, [C. Shen](#), L. Wang, A. van den Hengel, C. Wang (2014), “[Encoding High Dimensional Local Features by Sparse Coding Based Fisher Vectors](#)”, *Advances in Neural Information Processing Systems (NeurIPS'14)*.
107. [C. Shen](#), G. Lin, A. van den Hengel (2014), “[StructBoost: Boosting Methods for Predicting Structured Output Variables](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
106. L. Wang, L. Zhou, [C. Shen](#), L. Liu, H. Liu (2014), “[A Hierarchical Word-Merging Algorithm With Class Separability Measure](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
105. Y. Lu, L. Wang, J. Lu, J. Yang, [C. Shen](#) (2014), “[Multiple Kernel Clustering Based on Centered Kernel Alignment](#)”, *Pattern Recognition*.
104. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2014), “[Large-Margin Learning of Compact Binary Image Encodings](#)”, *IEEE Trans. Image Processing*.
103. Y. Yan, [C. Shen](#), H. Wang (2014), “[Efficient Semidefinite Spectral Clustering Via Lagrange Duality](#)”, *IEEE Trans. Image Processing*.
102. F. Shen, [C. Shen](#), R. Hill, A. van den Hengel, Z. Tang (2014), “[Fast Approximate L₁ Minimization: Speeding up Robust Regression](#)”, *Computational Statistics & Data Analysis*.

101. F. Liu, L. Zhou, [C. Shen](#), J. Yin (2014), “[Multiple Kernel Learning in the Primal for Multi-Modal Alzheimer'S Disease Classification](#)” [Online published at IEEE: 10 October 2013], *IEEE J. Biomedical & Health Informatics*.
100. Y. Li, W. Jia, [C. Shen](#), A. van den Hengel (2014), “[Characterness: An Indicator of Text in the Wild](#)”, *IEEE Trans. Image Processing*.
99. [C. Shen](#), J. Kim, F. Liu, L. Wang, A. van den Hengel (2014), “[Efficient Dual Approach to Distance Metric Learning](#)”, *IEEE Trans. Neural Networks & Learning Systems*.
98. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2014), “[A Scalable Stage-Wise Approach to Large-Margin Multi-Class Loss Based Boosting](#)”, *IEEE Trans. Neural Networks & Learning Systems*.
97. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2014), “[Asymmetric Pruning for Learning Cascade Detectors](#)”, *IEEE Trans. Multimedia*.
96. X. Li, W. Hu, [C. Shen](#), A. Dick, Z. Zhang (2014), “[Context-Aware Hypergraph Construction for Robust Spectral Clustering](#)”, *IEEE Trans. Knowledge & Data Engineering*.
95. S. Paisitkriangkrai, [C. Shen](#), Q. Shi, A. van den Hengel (2014), “[RandomBoost: Simplified Multi-Class Boosting Through Randomization](#)”, *IEEE Trans. Neural Networks & Learning Systems*.
94. F. Shen, [C. Shen](#), Q. Shi, A. van den Hengel, Z. Tang (2013), “[Inductive Hashing on Manifolds](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'13)*.
93. X. Li, [C. Shen](#), A. Dick, A. van den Hengel (2013), “[Learning Compact Binary Codes for Visual Tracking](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'13)*.
92. Z. Wang, Q. Shi, [C. Shen](#), A. van den Hengel (2013), “[Bilinear Programming for Human Activity Recognition With Unknown MRF Graphs](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'13)*.
91. P. Wang, [C. Shen](#), A. van den Hengel (2013), “[A Fast Semidefinite Approach to Solving Binary Quadratic Problems](#)” [Oral presentation, 60 out of 1870 submissions], *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'13)*.
90. R. Yao, Q. Shi, [C. Shen](#), Y. Zhang, A. van den Hengel (2013), “[Part-Based Visual Tracking With Online Latent Structural Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'13)*.
89. G. Lin, [C. Shen](#), D. Suter, A. van den Hengel (2013), “[A General Two-Step Approach to Learning-Based Hashing](#)”, *IEEE Int. Conf. Computer Vision (ICCV'13)*.
88. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2013), “[Efficient Pedestrian Detection by Directly Optimizing the Partial Area Under the ROC Curve](#)”, *IEEE Int. Conf. Computer Vision (ICCV'13)*.
87. X. Li, Y. Li, [C. Shen](#), A. Dick, A. van den Hengel (2013), “[Contextual Hypergraph Modeling for Salient Object Detection](#)”, *IEEE Int. Conf. Computer Vision (ICCV'13)*.
86. M. Harandi, C. Sanderson, [C. Shen](#), B. Lovell (2013), “[Dictionary Learning and Sparse Coding on Grassmann Manifolds: An Extrinsic Solution](#)”, *IEEE Int. Conf. Computer Vision (ICCV'13)*.
85. X. Li, G. Lin, [C. Shen](#), A. van den Hengel, A. Dick (2013), “[Learning Hash Functions Using Column Generation](#)” [Oral presentation], *Int. Conf. Machine Learning (ICML'13)*.
84. G. Lin, [C. Shen](#), A. van den Hengel (2013), “[Approximate Constraint Generation for Efficient Structured Boosting](#)”, *IEEE Conf. Image Processing (ICIP'13)*.
83. Y. Li, [C. Shen](#), W. Jia, A. van den Hengel (2013), “[Leveraging Surrounding Context for Scene Text Detection](#)”, *IEEE Conf. Image Processing (ICIP'13)*.
82. C. Zhang, J. Bastian, [C. Shen](#), A. van den Hengel, T. Shen (2013), “[Extended Depth-of-Field Via Focus Stacking and Graph Cuts](#)”, *IEEE Conf. Image Processing (ICIP'13)*.
81. X. Li, A. Dick, [C. Shen](#), A. van den Hengel, H. Wang (2013), “[Incremental Learning of 3D-DCT Compact Representations for Robust Visual Tracking](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.

80. [C. Shen](#), P. Wang, S. Paisitkriangkrai, A. van den Hengel (2013), “[Training Effective Node Classifiers for Cascade Classification](#)”, *Int. J. Computer Vision*.
79. [C. Shen](#), H. Li, A. van den Hengel (2013), “[Fully Corrective Boosting With Arbitrary Loss and Regularization](#)”, *Neural Networks*.
78. X. Li, W. Hu, [C. Shen](#), Z. Zhang, A. Dick, A. van den Hengel (2013), “[A Survey of Appearance Models in Visual Object Tracking](#)”, *ACM Trans. Intelligent Systems & Technology*.
77. X. Li, A. Dick, [C. Shen](#), Z. Zhang, A. van den Hengel, H. Wang (2013), “[Visual Tracking With Spatio-Temporal Dempster-Shafer Information Fusion](#)”, *IEEE Trans. Image Processing*.
76. F. Shen, [C. Shen](#), A. van den Hengel, Z. Tang (2013), “[Approximate Least Trimmed Sum of Squares Fitting and Applications in Image Analysis](#)”, *IEEE Trans. Image Processing*.
75. L. Luo, [C. Shen](#), C. Zhang, A. van den Hengel (2013), “[Shape Similarity Analysis by Self-Tuning Locally Constrained Mixed-Diffusion](#)”, *IEEE Trans. Multimedia*.
74. X. Li, [C. Shen](#), Q. Shi, A. Dick, A. van den Hengel (2012), “[Non-Sparse Linear Representations for Visual Tracking With Online Reservoir Metric Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'12)*.
73. S. Paisitkriangkrai, [C. Shen](#), A. van den Hengel (2012), “[Sharing Features in Multi-Class Boosting Via Group Sparsity](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'12)*.
72. R. Yao, Q. Shi, [C. Shen](#), Y. Zhang, A. van den Hengel (2012), “[Robust Tracking With Weighted Online Structured Learning](#)”, *European Conf. Computer Vision (ECCV'12)*.
71. Q. Shi, [C. Shen](#), R. Hill, A. van den Hengel (2012), “[Is Margin Preserved After Random Projection?](#)” [This work provides an analysis of margin distortion under random projections, the conditions under which margins are preserved, and presents bounds on the margin distortion], *Int. Conf. Machine Learning (ICML'12)*.
70. G. Lin, [C. Shen](#), A. van den Hengel, D. Suter (2012), “[Fast Training of Effective Multi-Class Boosting Using Coordinate Descent Optimization](#)”, *Asian Conf. Computer Vision (ACCV'12)*.
69. [C. Shen](#), P. Wang, F. Shen, H. Wang (2012), “[UBoost: Boosting With the Universum](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
68. [C. Shen](#), J. Kim, L. Wang, A. van den Hengel (2012), “[Positive Semidefinite Metric Learning Using Boosting-Like Algorithms](#)”, *J. Machine Learning Research*.
67. P. Wang, [C. Shen](#), N. Barnes, H. Zheng (2012), “[Fast and Robust Object Detection Using Asymmetric Totally-Corrective Boosting](#)”, *IEEE Trans. Neural Networks & Learning Systems*.
66. [C. Shen](#), A. van den Hengel (2012), “[Semidefinite Programming](#) (Book Chapter in: *Encyclopedia of Computer Vision*, Springer)”, 2012.
65. [C. Shen](#), Z. Hao (2011), “[A Direct Formulation for Totally-Corrective Multi-Class Boosting](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'11)*.
64. L. Liu, L. Wang, [C. Shen](#) (2011), “[A Generalized Probabilistic Framework for Compact Codebook Creation](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'11)*.
63. Q. Shi, A. Eriksson, A. van den Hengel, [C. Shen](#) (2011), “[Is Face Recognition Really a Compressive Sensing Problem?](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'11)*.
62. H. Li, [C. Shen](#), Q. Shi (2011), “[Real-Time Visual Tracking Using Compressive Sensing](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'11)*.
61. [C. Shen](#), J. Kim, L. Wang (2011), “[A Scalable Dual Approach to Semidefinite Metric Learning](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'11)*.
60. X. Li, A. Dick, H. Wang, [C. Shen](#), A. van den Hengel (2011), “[Graph Mode-Based Contextual Kernels for Robust SVM Tracking](#)”, *IEEE Int. Conf. Computer Vision (ICCV'11)*.

59. K. Park, [C. Shen](#), Z. Hao, J. Kim (2011), “[Efficiently Learning a Distance Metric for Large Margin Nearest Neighbor Classification](#)”, *AAAI Conf. Artificial Intelligence (AAAI'11)*.
58. L. Wang, [C. Shen](#), R. Hartley (2011), “[On the Optimality of Sequential Forward Feature Selection Using Class Separability Measure](#)”, *Int. Conf. Digital Image Computing: Techniques & Applications (DICTA'11)*.
57. T. Wang, X. He, [C. Shen](#), N. Barnes (2011), “[Laplacian Margin Distribution Boosting for Learning From Sparsely Labeled Data](#)”, *Int. Conf. Digital Image Computing: Techniques & Applications (DICTA'11)*.
56. [C. Shen](#), S. Paisitkriangkrai, J. Zhang (2011), “[Efficiently Learning a Detection Cascade With Sparse Eigenvectors](#)”, *IEEE Trans. Image Processing*.
55. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2011), “[Incremental Training of a Detector Using Online Sparse Eigen-Decomposition](#)”, *IEEE Trans. Image Processing*.
54. Q. Shi, H. Li, [C. Shen](#) (2010), “[Rapid Face Recognition Using Hashing](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'10)*.
53. [C. Shen](#), P. Wang, H. Li (2010), “[LACBoost And FisherBoost: Optimally Building Cascade Classifiers](#)”, *European Conf. Computer Vision (ECCV'10)*.
52. L. Zhou, L. Wang, [C. Shen](#), N. Barnes (2010), “[Hippocampal Shape Classification Using Redundancy Constrained Feature Selection](#)”, *Int. Conf. Medical Image Computing & Computer Assisted Intervention (MICCAI'10)*.
51. P. Wang, [C. Shen](#), N. Barnes, H. Zheng, Z. Ren (2010), “[Asymmetric Totally-Corrective Boosting for Real-Time Object Detection](#)” [Oral presentation], *Asian Conf. Computer Vision (ACCV'10)*.
50. Y. Zheng, [C. Shen](#), R. Hartley, X. Huang (2010), “[Pyramid Center-Symmetric Local Binary, Ternary Patterns for Effective Pedestrian Detection](#)”, *Asian Conf. Computer Vision (ACCV'10)*.
49. Z. Hao, [C. Shen](#), N. Barnes, B. Wang (2010), “[Totally-Corrective Multi-Class Boosting](#)”, *Asian Conf. Computer Vision (ACCV'10)*.
48. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2010), “[Face Detection With Effective Feature Extraction](#)”, *Asian Conf. Computer Vision (ACCV'10)*.
47. H. Li, P. Wang, [C. Shen](#) (2010), “[Robust Face Recognition Via Accurate Face Alignment and Sparse Representation](#)”, *Int. Conf. on Digital Image Computing: Techniques & Applications (DICTA'10)*.
46. W. Wang, J. Zhang, [C. Shen](#) (2010), “[Improved Human Detection and Classification in Thermal Images](#)”, *IEEE Int. Conf. Image Processing (ICIP'10)*.
45. P. Wang, [C. Shen](#), H. Zheng, Z. Ren (2010), “[Training a Multi-Exit Cascade With Linear Asymmetric Classification for Efficient Object Detection](#)”, *IEEE Int. Conf. Image Processing (ICIP'10)*.
44. [C. Shen](#), H. Li (2010), “[On the Dual Formulation of Boosting Algorithms](#)”, *IEEE Trans. Pattern Analysis & Machine Intelligence*.
43. [C. Shen](#), J. Kim, L. Wang (2010), “[Scalable Large-Margin Mahalanobis Distance Metric Learning](#)”, *IEEE Trans. Neural Networks*.
42. L. Zhou, L. Wang, [C. Shen](#) (2010), “[Feature Selection With Redundancy-Constrained Class Separability](#)”, *IEEE Trans. Neural Networks*.
41. [C. Shen](#), H. Li (2010), “[Boosting Through Optimization of Margin Distributions](#)”, *IEEE Trans. Neural Networks*.
40. [C. Shen](#), J. Kim, H. Wang (2010), “[Generalized Kernel-Based Visual Tracking](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
39. H. Li, [C. Shen](#) (2010), “[Interactive Color Image Segmentation With Linear Programming](#)”, *Machine Vision & Applications*.
38. J. Zhang, [C. Shen](#), G. Geers, Q. Wu (2010), “[Proceedings of International Conference on Digital Image Computing: Techniques and Applications](#)”, 2010.

37. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2009), “[Efficiently Training a Better Visual Detector With Sparse Eigenvectors](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'09)*.
36. [C. Shen](#), J. Kim, L. Wang, A. van den Hengel (2009), “[Positive Semidefinite Metric Learning With Boosting](#)”, *Advances in Neural Information Processing Systems (NeurIPS'09)*.
35. Y. Dai, H. Li, M. He, [C. Shen](#) (2009), “[Smooth Approximation of \$L_1\$ -Norm for Multi-View Geometry](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'09)*.
34. W. Wang, [C. Shen](#), J. Zhang, S. Paisitkriangkrai (2009), “[A Two-Layer Night-Time Vehicle Detector](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'09)*.
33. P. Wang, [C. Shen](#), H. Zheng, Z. Ren (2009), “[A Variant of the Trace Quotient Formulation for Dimensionality Reduction](#)”, *9th Asian Conf. Computer Vision (ACCV'09)*.
32. J. Kim, [C. Shen](#), L. Wang (2009), “[A Scalable Algorithm for Learning a Mahalanobis Distance Metric](#)”, *9th Asian Conf. Computer Vision (ACCV'09)*.
31. L. Wang, L. Zhou, [C. Shen](#) (2008), “[A Fast Algorithm for Creating a Compact and Discriminative Visual Codebook](#)”, *European Conf. Computer Vision (ECCV'08)*.
30. [C. Shen](#), A. Welsh, L. Wang (2008), “[PSDBOost: Matrix-Generation Linear Programming for Positive Semidefinite Matrices Learning](#)”, *Advances in Neural Information Processing Systems (NeurIPS'08)*.
29. Y. Lu, L. Wang, R. Hartley, H. Li, [C. Shen](#) (2008), “[Multi-View Human Motion Capture With an Improved Deformation Skin Model](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'08)*.
28. J. Kim, [C. Shen](#), L. Wang (2008), “[Learning Cascaded Reduced-Set SVMs Using Linear Programming](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'08)*.
27. H. Li, [C. Shen](#) (2008), “[Boosting the Minimum Margin: LPBoost vs. AdaBoost](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'08)*.
26. [C. Shen](#), H. Li, M.J. Brooks (2008), “[Self-Calibrating Cameras Using Semidefinite Programming](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'08)*.
25. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2008), “[Real-Time Pedestrian Detection Using a Boosted Multi-Layer Classifier](#)”, *8th IEEE Int. Workshop on Visual Surveillance, in conjunction with European Conf. Computer Vision (ECCVW'08)*.
24. [C. Shen](#), S. Paisitkriangkrai, J. Zhang (2008), “[Face Detection From Few Training Examples](#)”, *IEEE Int. Conf. Image Processing (ICIP'08)*.
23. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2008), “[Performance Evaluation of Local Features in Human Classification and Detection](#)” [Invited submission, special issue of DICTA2007], *IET Computer Vision*.
22. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2008), “[Fast Pedestrian Detection Using a Cascade of Boosted Covariance Features](#)”, *IEEE Trans. Circuits & Systems for Video Technology*.
21. [C. Shen](#), H. Li, M.J. Brooks (2008), “[Supervised Dimensionality Reduction Via Sequential Semidefinite Programming](#)”, *Pattern Recognition*.
20. Q. Nguyen, A. Robles-Kelly, [C. Shen](#) (2007), “[Kernel-Based Tracking From a Probabilistic Viewpoint](#)”, *IEEE Conf. Computer Vision & Pattern Recognition (CVPR'07)*.
19. H. Li, [C. Shen](#), Z. Wen (2007), “[Color Image Labelling Using Linear Programming](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'07)*.
18. S. Paisitkriangkrai, [C. Shen](#), J. Zhang (2007), “[An Experimental Evaluation of Local Features for Pedestrian Classification](#)” [Best Paper Award], *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'07)*.
17. [C. Shen](#), H. Li, M.J. Brooks (2007), “[Feature Extraction Using Sequential Semidefinite Programming](#)”, *Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'07)*.

16. [C. Shen](#), H. Li, M.J. Brooks (2007), “[A Convex Programming Approach to the Trace Quotient Problem](#)”, 8th Asian Conf. Computer Vision (ACCV'07).
15. H. Li, [C. Shen](#) (2007), “[Object-Respecting Colour Image Segmentation: An LP Approach](#)”, IEEE Int. Conf. Image Processing (ICIP'07).
14. H. Wang, D. Suter, K. Schindler, [C. Shen](#) (2007), “[Adaptive Object Tracking Based on an Effective Appearance Filter](#)” [Featured article of September issue 2007], IEEE Trans. Pattern Analysis & Machine Intelligence.
13. [C. Shen](#), M.J. Brooks, A. van den Hengel (2007), “[Fast Global Kernel Density Mode Seeking: Applications to Localization and Tracking](#)”, IEEE Trans. Image Processing.
12. [C. Shen](#), H. Li, M.J. Brooks (2006), “[Classification-Based Likelihood Functions for Bayesian Tracking](#)”, IEEE Int. Conf. Advanced Video & Signal based Surveillance (AVSS'06).
11. H. Li, [C. Shen](#) (2006), “[An LMI Approach for Reliable PTZ Camera Self-Calibration](#)”, IEEE Int. Conf. Advanced Video & Signal based Surveillance (AVSS'06).
10. Q. Nguyen, A. Robles-Kelly, [C. Shen](#) (2006), “[Enhanced Kernel-Based Tracking for Monochromatic and Thermographic Video](#)”, IEEE Int. Conf. Advanced Video & Signal based Surveillance (AVSS'06).
9. [C. Shen](#), M.J. Brooks, A. van den Hengel (2005), “[Fast Global Kernel Density Mode Seeking With Application to Localisation and Tracking](#)” [Oral presentation, 45 out of 1200 submissions], IEEE Int. Conf. Computer Vision (ICCV'05).
8. [C. Shen](#), M.J. Brooks (2005), “[Adaptive Over-Relaxed Mean Shift](#)” [Errata: in figure 3 square marker and circle marker should be swapped], 8th Int. Symposium on Signal Processing & Its Applications (ISSPA'05).
7. [C. Shen](#), A. van den Hengel, M.J. Brooks (2005), “[Visual Tracking Via Efficient Kernel Discriminant Subspace Learning](#)”, IEEE Int. Conf. Image Processing (ICIP'05).
6. [C. Shen](#), M.J. Brooks, A. van den Hengel (2005), “[Augmented Particle Filtering for Efficient Visual Tracking](#)”, IEEE Int. Conf. Image Processing (ICIP'05).
5. [C. Shen](#), A. van den Hengel, A. Dick, M.J. Brooks (2004), “[2D Articulated Tracking With Dynamic Bayesian Networks](#)”, Int. Conf. Computer & Information Technology (CIT'04).
4. [C. Shen](#), A. van den Hengel, A. Dick, M.J. Brooks (2004), “[Enhanced Importance Sampling: Unscented Auxiliary Particle Filtering for Visual Tracking](#)”, Australian Joint Conf. Artificial Intelligence (AI'04).
3. Z. Lin, J. Lu, [C. Shen](#), X. Qiu, B. Xu (2004), “[Active Control of Radiation From a Piston Set in a Rigid Sphere](#)”, J. Acoustical Society of America.
2. [C. Shen](#), A. van den Hengel, A. Dick (2003), “[Probabilistic Multiple Cue Integration for Particle Filter Based Tracking](#)” [Nominated for Best Student Paper Award], Int. Conf. Digital Image Computing - Techniques & Applications (DICTA'03).
1. J. Lu, [C. Shen](#), X. Qiu, B. Xu (2003), “[Lattice Form Adaptive Infinite Impulse Response Filtering Algorithm for Active Noise Control](#)”, J. Acoustical Society of America.

Generated 17 May 2026, 22:29